

Name: _____

GCSE Biology Unit 2 Foundation Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /80	Grade	Good Questions	Bad Questions	Reasons
2018					
2019					
2022					
2023					

Grade boundaries

Year	G	F	E	D	C	Max UMS
2018	8	13	18	24	30	36
2019	7	13	19	25	31	37
2022	13	18	23	29	35	41
2023	12	18	24	30	37	44

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3400U20-1



BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

FOUNDATION TIER

TUESDAY, 15 MAY 2018 – AFTERNOON
1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **6(a)** is a quality of extended response (QER) question where your writing skills will be assessed.



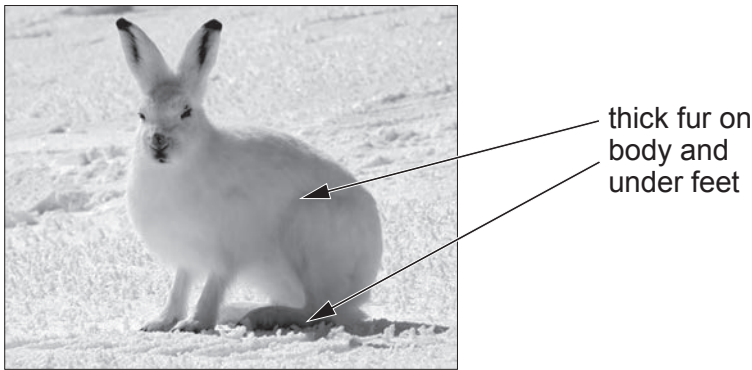
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For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	14	
3.	8	
4.	7	
5.	12	
6.	8	
7.	7	
8.	13	
Total	80	

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Answer all questions

1. The photograph shows an arctic hare (*Lepus arcticus*).



(a) The table below shows some classes of vertebrate animals.

Class	Features of animals	Examples
fish	skin with scales	goldfish, cod
reptiles	skin with scales	crocodiles, snakes
birds	skin with feathers	eagle, pigeon
mammals	skin with hair	horse, cat

(i) Use the information above and your own knowledge to complete the classification of the arctic hare. [3]

Kingdom animal

Phylum vertebrate

Class

Genus

Species

(ii) Why do scientists use scientific names for living organisms? [1]

.....



(b) Read the information about the arctic hare and use it to answer the questions which follow.

The arctic hare lives in northern countries where the climate is very cold. The average life expectancy is five years. Each has a body mass of 5 – 7 kg and 20% of this is a layer of fat under the skin, which helps to reduce heat loss. They live in large groups of about 200 individuals and huddle together while sleeping, to retain heat.

Arctic hares feed on berries, twigs and moss which they can dig out of snow with their strong feet. Only a few other species eat the same food as the hares but many others, such as foxes, wolves and lynx kill and eat large numbers of hares.

(i) Complete the table below by writing ‘true’ or ‘false’ for each of the statements about the arctic hare. [3]

Statement	True or False
All arctic hares live for at least five years.	
Arctic hares have little inter-specific competition for food.	
The population size of arctic hares is affected by predation.	
The layer of fat under the skin raises the body temperature.	

(ii) Explain why arctic hares are at a high risk of spreading disease while they sleep. [2]

.....

.....

(c) Give **two** reasons why the arctic hare’s coat increases its chances of survival. [2]

.....

.....



2. The concentration of glucose in the blood is normally between 60 mg and 160 mg per 100 cm³ of blood. Insulin reduces the concentration and prevents it from rising above the normal level.

(a) State the name of the organ which produces insulin. [1]

.....

(b) Doctors investigated the concentration of blood glucose in Kate and John. They suspected that Kate had diabetes but they knew that John did not.

Their blood glucose was measured after taking a glucose drink.
The results for John are shown in the table below. Kate's results are shown on the graph.

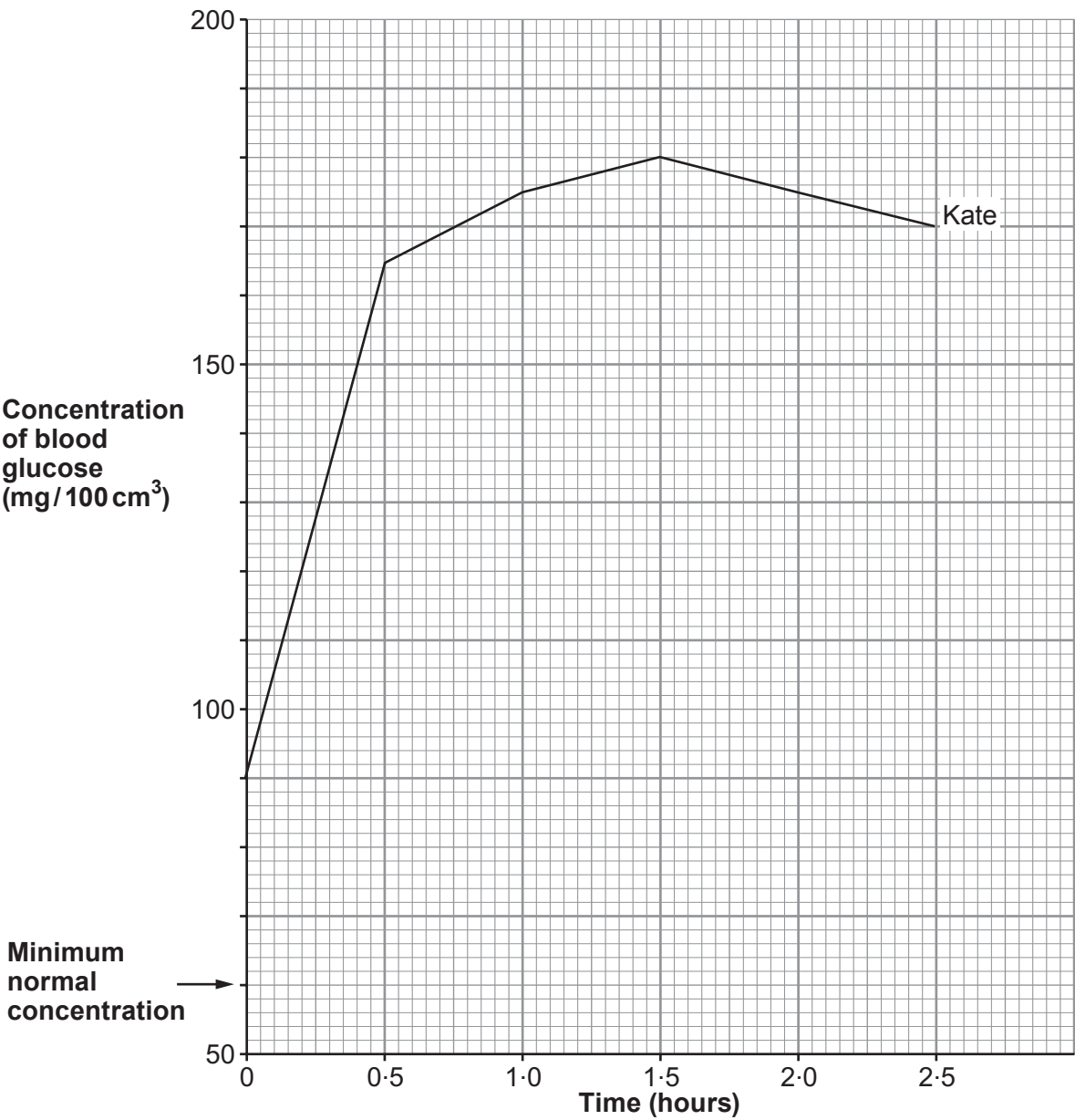
Time (hours)	John's blood glucose level (mg / 100 cm ³)
0	80
0.5	115
1.0	134
1.5	110
2.0	95
2.5	84

(i) Complete the graph of results by: [4]

- I. drawing an arrow on the axis for glucose concentration to show the **maximum normal concentration** of blood glucose (the minimum has been done for you);
- II. plotting the blood glucose results for John;
- III. joining your plots with a ruler and labelling your line.



Graph of results



Use the graph to answer the questions.

- (ii) At what time does insulin start to affect the concentration of glucose in John's blood? Give a reason for your answer. [2]

.....

.....



Examiner
only

(iii) How do the results for Kate at 0.5 hours show that she has diabetes? [1]

.....

(iv) Describe how the results for Kate are different from those of John, between 0.5 and 2 hours. [3]

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(v) The doctors concluded that Kate had diabetes. How could they increase the confidence they had in their results? [1]

.....

(c) (i) State **one** way in which Kate’s diabetes could be treated. [1]

.....

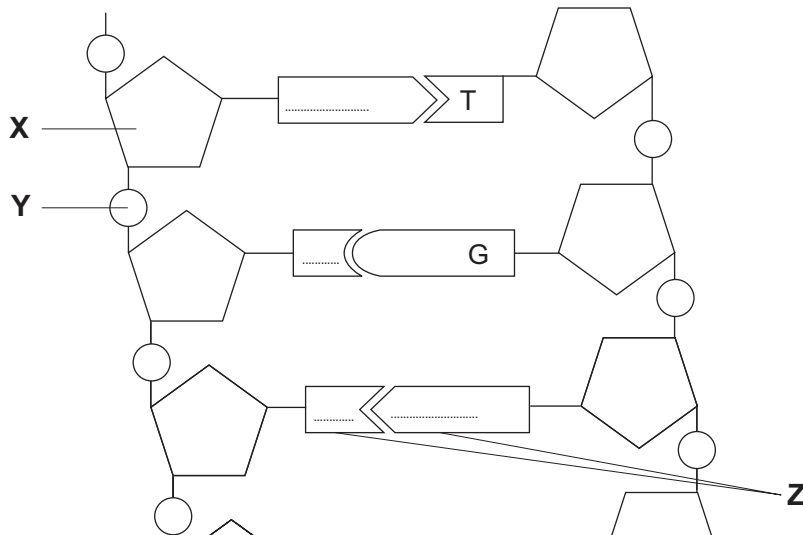
(ii) Arthur is 70 years old. He produces insulin but his liver cells do not respond to it. State the precise name of this medical condition. [1]

.....

14



3. (a) The diagram below shows a small section of DNA.



- (i) Which one of the following pairs of substances correctly describes labels **X** and **Y** on the diagram?

Underline your answer

[1]

salt and sugar

sugar and phosphate

acid and phosphate

salt and acid

- (ii) The structures labelled **Z** are bases.

Complete the letter names for the **four** missing bases on the diagram.

[2]

- (iii) State why the order of the bases in DNA is important in the production of proteins.

[2]

.....

.....



(b) Apart from identical twins, no two persons have identical DNA. Samples of DNA can be analysed to produce DNA profiles which can be used to identify individuals in criminal investigations.

The diagram below shows five DNA profiles.

DNA from crime scene	DNA from suspects			
	suspect 1	suspect 2	suspect 3	suspect 4
████████	████████	████████	████████	████████
████████	████████		████████	████████
████████		████████	████████	
████████	████████	████████	████████	
	████████	████████		████████
████████	████████	████████	████████	████████
████████	████████		████████	████████
████████	████████	████████	████████	████████
████████			████████	
████████	████████	████████	████████	████████

(i) From the diagram, identify the suspect whose DNA was found at the crime scene. Give a reason for your answer. [1]

Suspect

Reason

(ii) Apart from criminal investigations, state **one other** use of DNA profiling. [1]

.....

(iii) After DNA profiles have been used in investigations they are often retained for future reference. Suggest **one** reason why some people may object to their DNA profiles being retained by the police. [1]

.....



4. In 1825, plant collectors brought Japanese knotweed (*Fallopia japonica*) into the UK. It spread into many habitats, mostly near rivers. It is now out of control in most areas, eliminating other plant species and damaging roads and buildings.



Japanese knotweed in summer



sap-sucking louse

Japanese knotweed grows rapidly in summer. Plants reach 4 metres in height and underground stems grow to 25 metres in length.

Scientists working for the Welsh government investigated the use of a sap-sucking louse (*Aphalara itadori*), to destroy Japanese knotweed in a number of trials in parts of the UK.

In the trials, the louse reduced the growth of Japanese knotweed by 60%. The louse did not harm any other species and reproduced quickly in summer. Most of the lice, however, died in the winter.

- (a) (i) Which of the following describes Japanese knotweed in the UK? Write the correct letter in the box. [1]

- A an endangered alien species
- B an alien invasive species
- C an endangered native species
- D a native invasive species

answer

- (ii) How does Japanese knotweed affect biodiversity in the areas where it grows in the UK? Give a reason for your answer. [1]

.....

.....



Examiner
only

(b) (i) State the scientific term used when an organism is used to destroy a pest species. [1]

(ii) Calculate the length of underground stems produced in Japanese knotweed when the sap-sucking louse is present. [2]

length = m

(c) (i) Following the trials, the scientists concluded that the sap-sucking louse was effective against Japanese knotweed as it reduced growth by 60%.

They also decided that it would be suitable to use this method on a wider scale throughout the UK. Give **one** piece of evidence which supports this decision. [1]

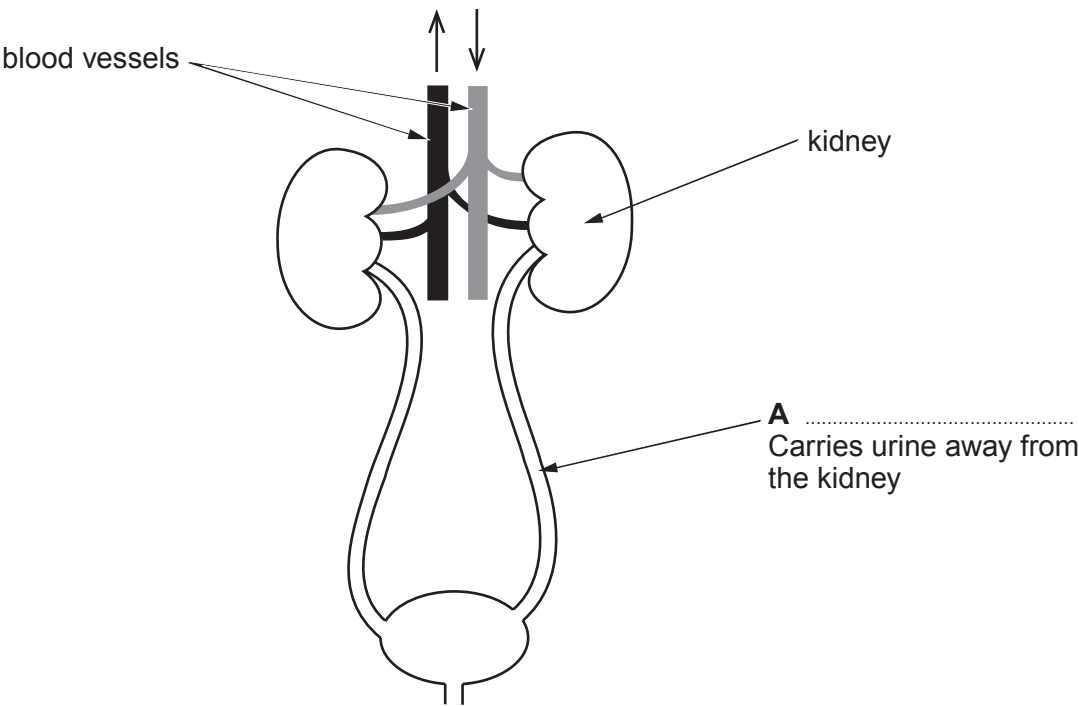
(ii) How could these scientists check that the results were reproducible? [1]

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5. The diagram shows the human excretory system.



- (a) **Complete** label **A** on the diagram. [1]
- (b) A doctor investigated the concentrations of some substances present in the blood entering and leaving a patient's kidneys.

Substance	Concentration in blood (a.u.)	
	blood entering kidney	blood leaving kidney
glucose	168	168
salt (sodium)	35	33
protein	180	150
urea	314	11

- (i) From the table, state the waste substance which would be present in the patient's urine at the highest concentration. [1]

.....



- (ii) The doctor thought that this patient’s urine would contain protein but no glucose. What is the evidence in the table to support this? [2]

.....

.....

.....

- (iii) The urine was tested to find out if the doctor was correct.

I. Protein test

State the name of the chemical reagent added to the urine to test for the presence of protein and the colour which would indicate a positive result. [2]

.....

.....

II. Glucose test

When a sample of urine was tested for glucose a chemical reagent was added and the mixture was heated strongly. The colour remained blue showing the doctor to be correct.

Give the name of this chemical reagent and explain how the result would be different if the doctor was incorrect. [2]

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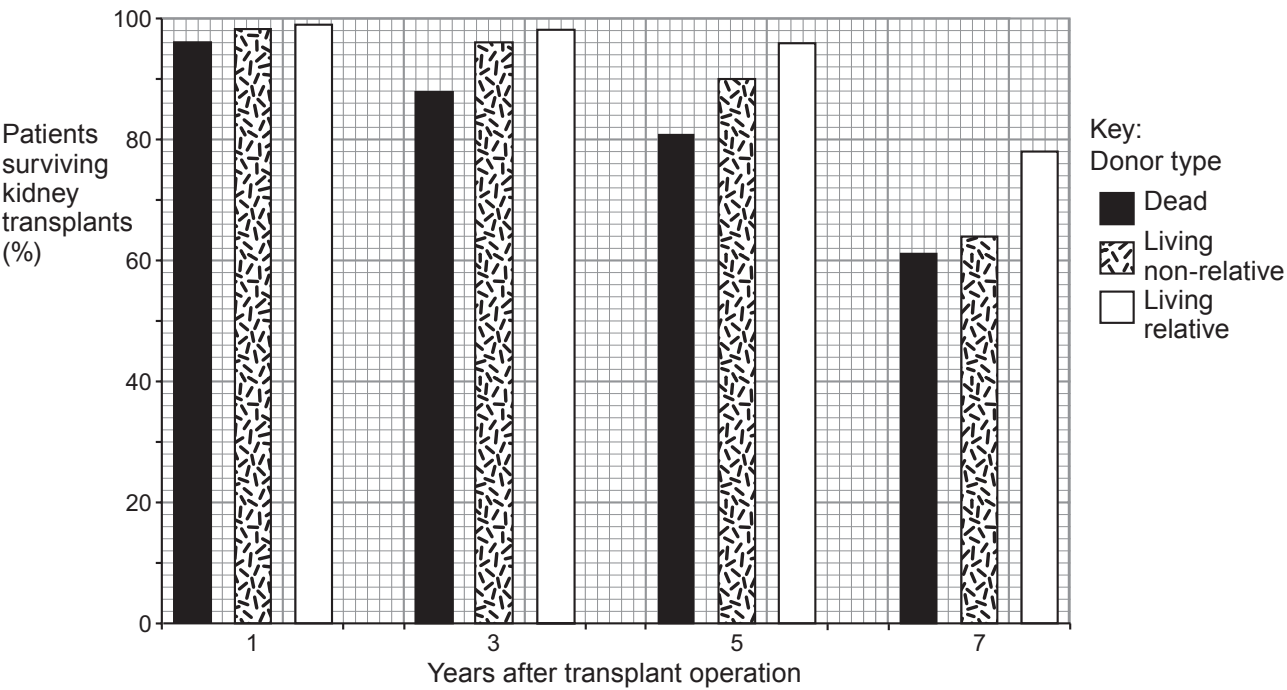
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(c) In the case of serious kidney disease a transplant operation can be carried out if a suitable kidney is available from a donor.

The bar chart shows the results of kidney transplant operations with donors of different types. All the recipients and donors were aged 30 – 50 years.



From the bar chart state **three** conclusions that can be drawn to compare the survival of patients after kidney transplant operations with different types of donors. [3]

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(d) Insufficient donor kidneys are available. What other form of treatment can be offered to people with kidney disease? [1]

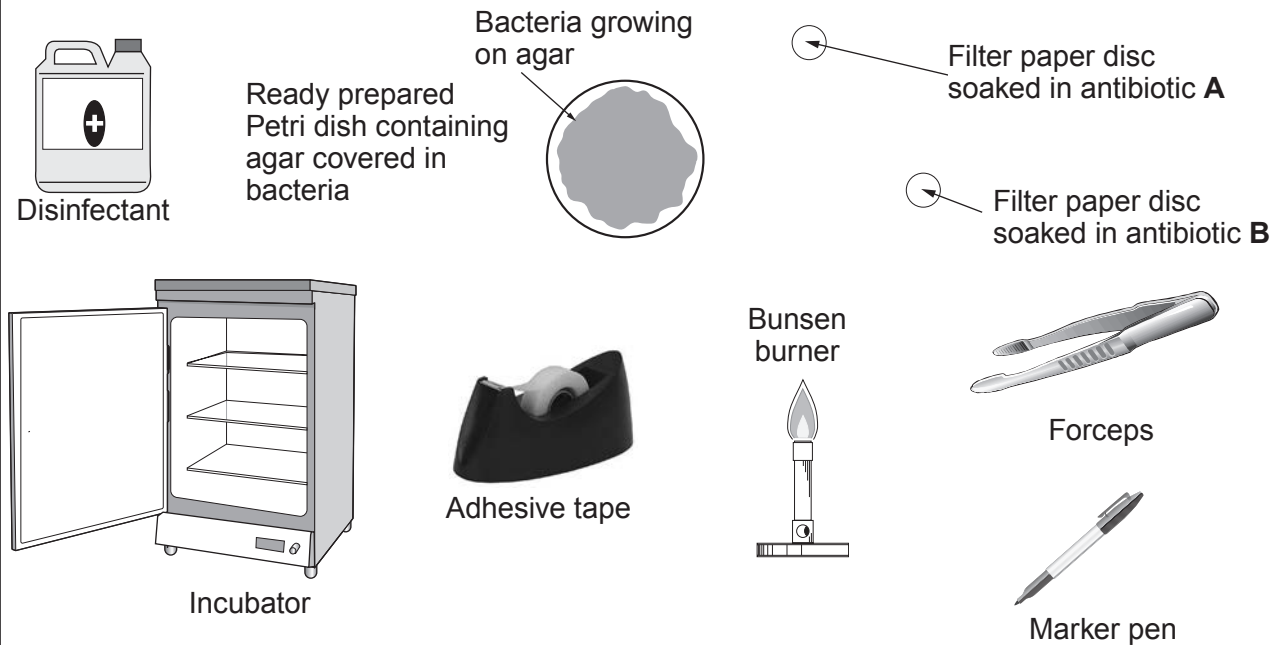
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6. (a) Describe how you would investigate the effects of two different antibiotics **A** and **B**, on the growth of bacteria. Include examples of aseptic technique in your answer. [6 QER]

You are provided with the following apparatus.



(b) The bacterium MRSA is resistant to most antibiotics. State **two** ways in which the spread of MRSA can be reduced. [2]

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Examiner
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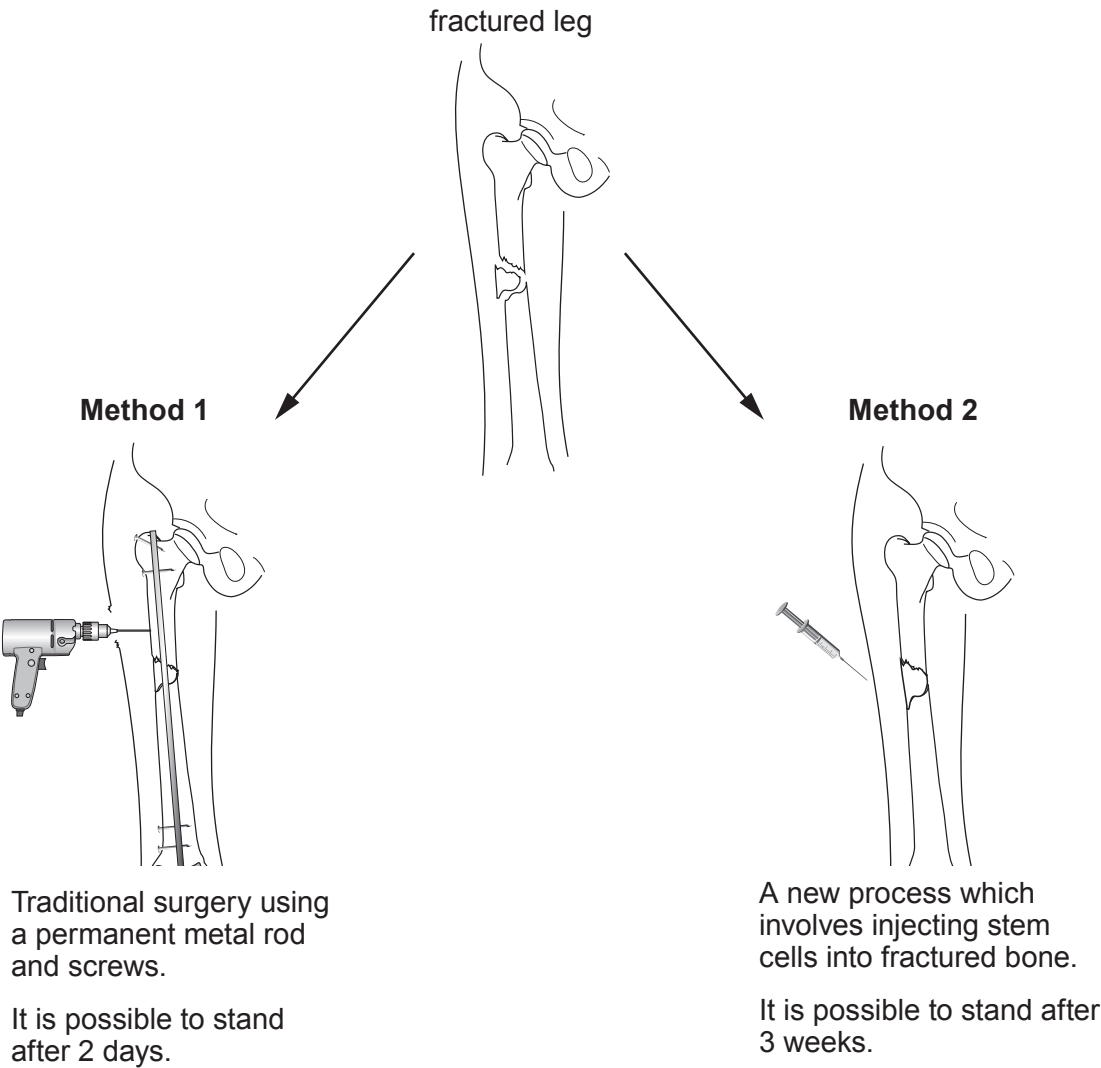
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7. The photograph below shows a young ice skater, Tracey, who falls and fractures her leg.



The diagram below shows two ways in which Tracey’s fractured leg could be treated in a large modern hospital.



Expected results of treatment

Time after treatment (weeks)	Percentage of bone healing (%)	
	Traditional surgery	Injection of stem cells
10	10	12
20	14	25
30	19	38
40	28	55
50	41	70
60	59	82

- (a) Give **one** reason why Tracey might choose traditional surgery and **one** reason why, alternatively, she might prefer to be treated by an injection of stem cells. [2]

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.....

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- (b) (i) When fractured bones heal, cells divide and multiply by mitosis.

Complete the table below.

[2]

Mitosis in human cells	
chromosomes in mother cell	46
number of daughter cells produced after one division	
number of chromosomes in daughter cells	

- (ii) Following mitosis, what must happen to stem cells, in order for them to repair the fractured bone? [1]

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(iii) State the name of the disease that can occur if cell division by mitosis is uncontrolled. [1]

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(c) Stem cells can be obtained from both adults and embryos.

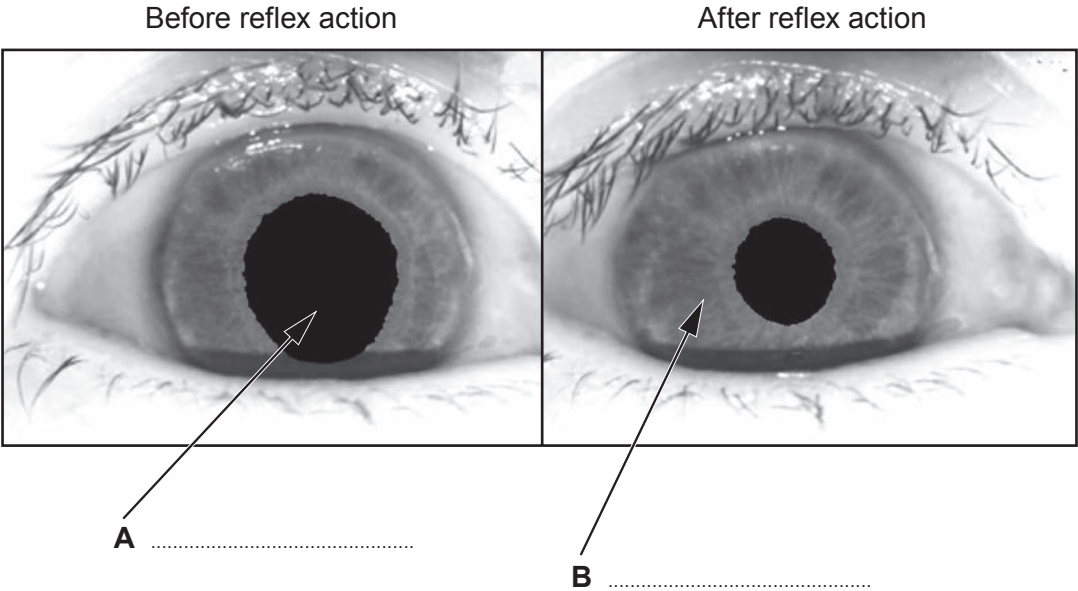
Give **one** reason why some people have a strong personal objection to the use of embryonic stem cells in medical research. [1]

.....

7



8. The photographs below show the appearance of an eye before and after a reflex action which occurs in response to a change in light intensity.



- (a) (i) Label **A** and **B** on the diagram. [1]
- (ii) From the photographs, describe how **and** why parts **A** and **B** of the eye alter when the light intensity changes. [3]

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- (iii) State **two** features of **all** reflex actions. [1]

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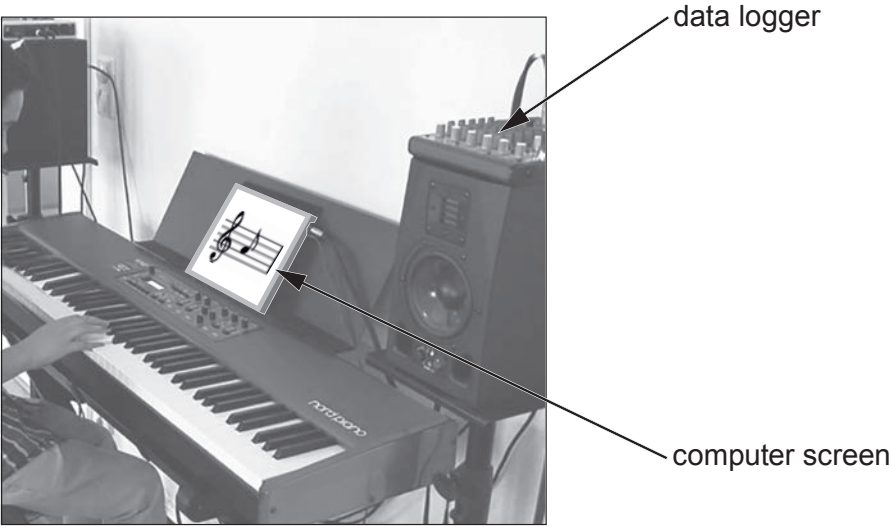


(b) Josie investigated reaction time in humans.

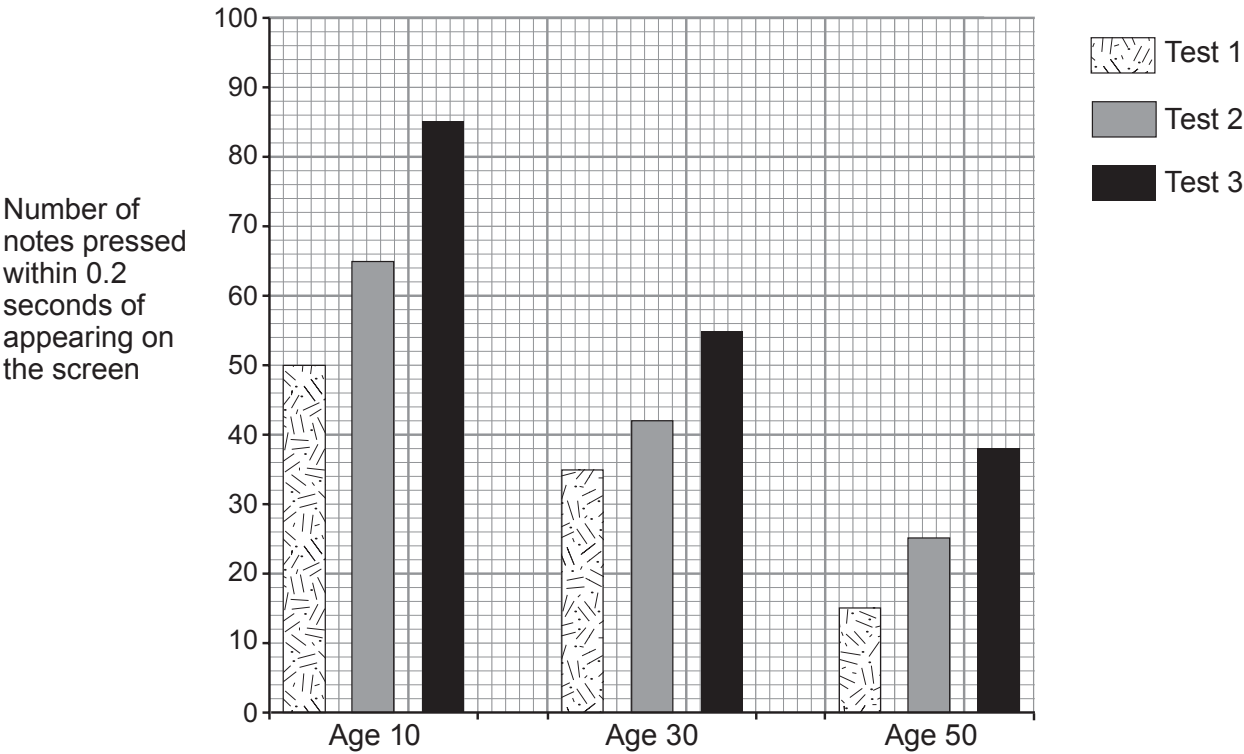
She tested three people, two males and one female of ages 10, 30 and 50 years old. They had between one and 20 years experience of playing the keyboard.

By means of a computer app, 90 random music notes flashed one by one onto a screen. The person being tested then instantly pressed each note on the keyboard as soon as it was seen. Each person did the test three times. No incorrect notes were pressed.

A data logger recorded the number of notes which were pressed within 0.2 seconds of appearing on the screen.



The bar chart shows the results of the investigation.



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Use the bar chart to answer the questions.

(i) How does repeating the test affect reaction time? [1]

.....

(ii) Calculate the percentage change between tests 1 and 2 for age 50. Give your answer to one decimal place. [2]

Percentage change = %

(iii) From the data, what **two** conclusions could you make about the effects of age on reaction time? [2]

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(iv) I. Josie decided to try the investigation again and make it a fairer test of the effects of age. State **two** variables which she should control. [2]

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.....

II. State **one other** way in which the investigation could be improved. [1]

.....

.....

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END OF PAPER



Surname	Centre Number	Candidate Number
Other Names		0



GCSE

3400U20-1



BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

FOUNDATION TIER

TUESDAY, 14 MAY 2019 – AFTERNOON
1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	16	
3.	9	
4.	13	
5.	12	
6.	10	
7.	10	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **3(b)** is a quality of extended response (QER) question where your writing skills will be assessed.



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Answer all questions.

1. Read the information about obesity.

Fact file on Obesity

- In 2015, one in five adults and many children in the UK showed signs of being overweight or obese.
- Obesity makes it more likely that a person will develop medical conditions such as arthritis and type 2 diabetes as well as life-threatening diseases such as heart disease and cancer.
- There are more obese men than obese women but high blood pressure and type 2 diabetes are four times more common in obese women than in obese men.
- A diet low in saturated fat combined with regular exercise can be an effective way to reduce or avoid obesity.

(a) (i) Use the information to complete the table. Write **‘true’** or **‘false’** for each statement. [3]

Statement	True or False
In 2015, 20% of the population of the UK was obese.	
Obesity increases the risk of serious diseases.	
Obese women are more likely than obese men to have cells in their bodies which do not respond to insulin.	
A healthy lifestyle will reduce obesity.	

(ii) Complete the following sentence by selecting your answer from the words below. [1]

nerves blood bile

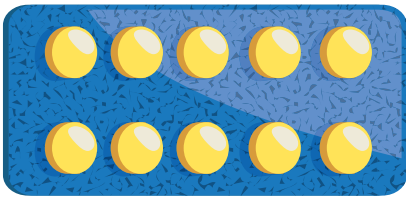
Hormones such as insulin are chemical messengers carried in the



(b) In 2010 a drug, Sibutramine, was developed to help treat obesity.

Ten patients in a hospital tested the drug for a month and doctors observed that it resulted in considerable weight loss.

Medical researchers then carried out a double-blind clinical trial, where 600 patients took the drug for two years. Patients in group **A** received Sibutramine tablets and patients in group **B** received a placebo.



Sibutramine tablets

(i) State **two** differences between the trial carried out by the medical researchers and the test carried out in the hospital. [2]

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(ii) State the meaning of the term *placebo*. [1]

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.....

(iii) Explain why a placebo is necessary in trials of new drugs. [1]

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- (iv) Choose the letter (**A-D**), which gives the meaning of the term double-blind trial. [1]
- A** Only the doctor knows whether the patient is in group A or group B
 - B** The doctor and the patient both know whether the patient is in group A or group B.
 - C** Neither the doctor nor the patient know whether the patient is in group A or group B.
 - D** Patients can choose whether they wish to be in group A or group B.

Answer

- (c) Before new medicines are tested on human patients they are usually tested on animals in laboratories. It is now possible to test some medicines using computer programs which can model the human body or use tissue cultures from human cells in Petri dishes.

State **one** advantage of using computer programs or tissue cultures rather than animals for testing medicines. [1]

.....

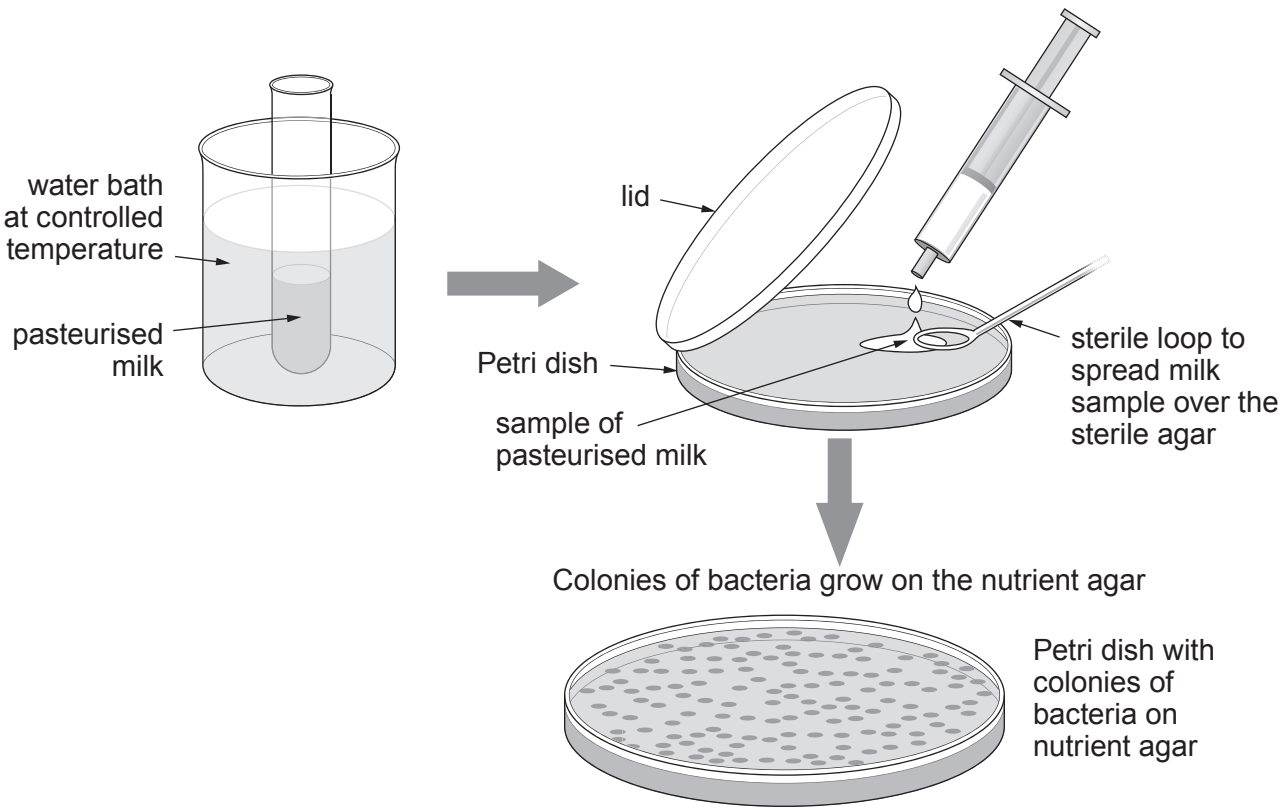
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2. Fresh milk contains bacteria. Milk sold by supermarkets is pasteurised and has been heated to 72 °C for 20 seconds to kill most of the bacteria before it is bottled and stored.

Students investigated the growth of bacteria in pasteurised milk which they stored at temperatures of 10, 25 and 35 °C. They took samples of the milk at regular intervals and placed them on sterile nutrient agar to compare the growth of bacteria.

The diagram shows their procedure.



- (a) (i) State why it would be important for each milk sample to be of the same volume. [1]
-
- (ii) Describe how the loop used for spreading the sample could be sterilised and why this sterilisation was important. [2]
-
-
-
-
- (iii) State what the students would do to the Petri dish to ensure aseptic conditions, after the milk sample had been spread on the agar. [1]
-



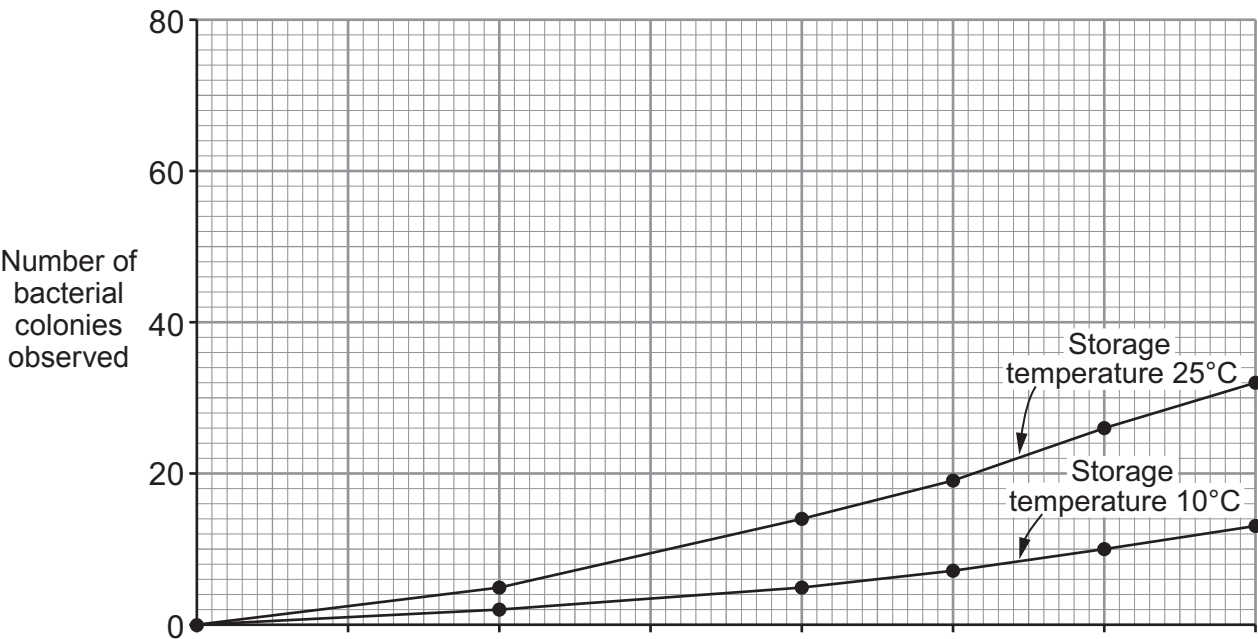
The results of the investigation are shown in the table and graph below.

Time of storage (hours)		Number of bacterial colonies observed		
		Storage temperature 10 °C	Storage temperature 25 °C	Storage temperature 35 °C
Start	0	0	0	0
	10	2	5	9
	20	5	14	28
	25	7	19	39
	30	10	26	50
	35	13	32	57

- (b) (i) Complete the graph by: [4]
- I. adding the scale and label for time;

II. plotting the results for 35 °C;

III. joining your plots with a ruler and labelling your line.



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(ii) Describe how the number of bacterial colonies changes over time and how this is affected by storage temperature. [2]

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.....

.....

.....

.....

(iii) I. Use the graph to estimate the difference in the number of colonies of bacteria between:
10 °C and 25 °C at 15 hours. [2]

Number of colonies

II. In the investigation, 50 colonies were found to have grown on the agar in a Petri dish. State how many bacteria would have been present in the sample of milk. Circle your answer. [1]

5 15 50 500

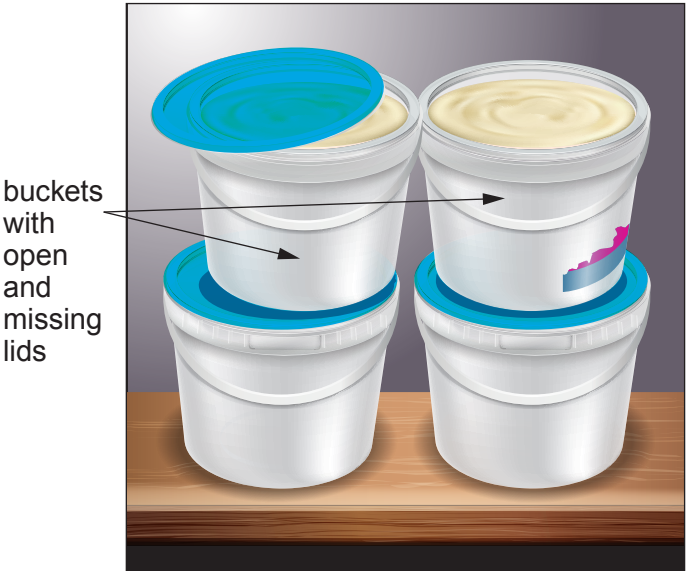
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(c) Food hygiene inspectors visited a restaurant in South Wales in the summer of 2010. They reported problems. The diagrams show some of their observations.

Buckets of yogurt made from pasteurised milk in a store cupboard at room temperature.

Close-up of a label on a bucket of yogurt.



From the problems they reported, the inspectors concluded that the level of hygiene at the restaurant was so low that it failed the inspection.

From the diagrams and your own knowledge explain the scientific reasons for the conclusion drawn by the inspectors. **Write your answers in the table below.** [3]

Problems reported by inspectors	Scientific reasons for inspectors' conclusion.
storage temperature	
lids of yogurt containers	
risk to customers' health	

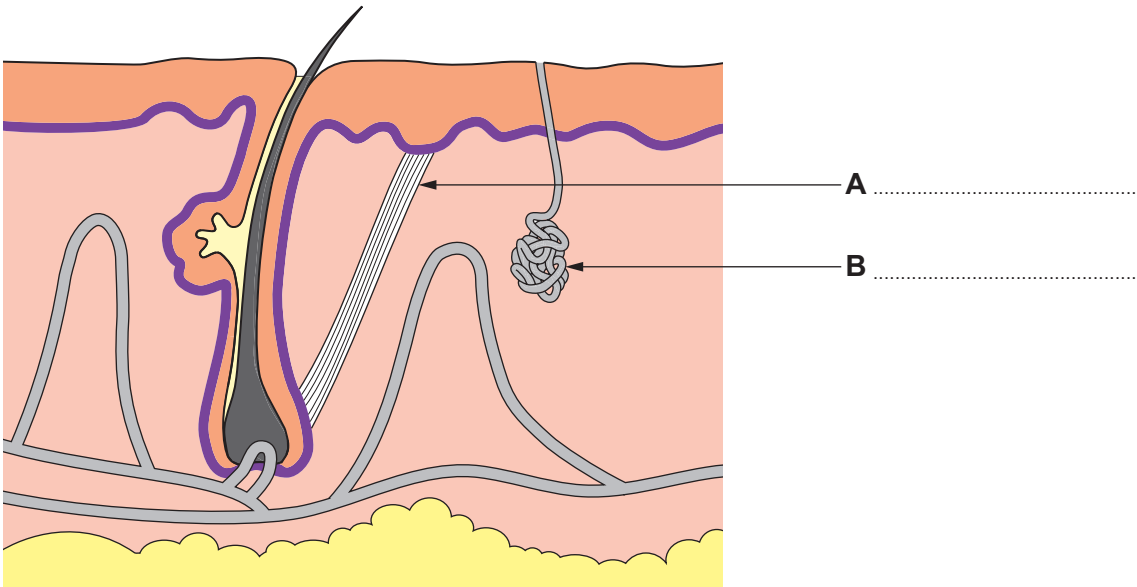


3. (a) (i) State how the skin defends the body against disease. [1]

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.....

The diagram below shows a section through human skin.



(ii) Label **A** and **B** on the diagram. [2]



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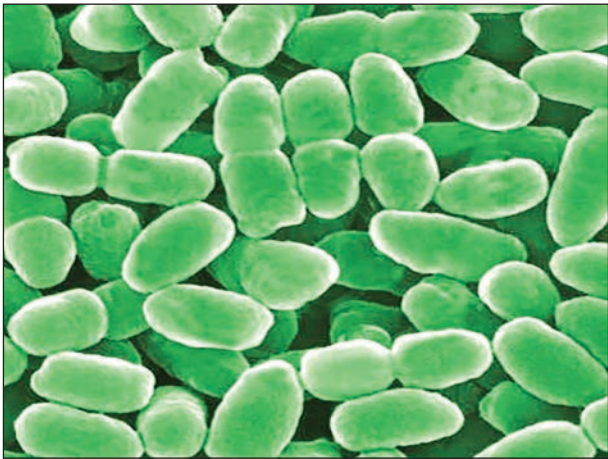
- (b) After playing rugby in warm conditions Steve undergoes a session in a cryotherapy chamber. This is a small room where the temperature is very low.



Describe and explain the changes that would occur in his skin in the cold conditions of the cryotherapy chamber. [6 QER]



4. Whooping Cough is a disease caused by the bacterium *Bordetella pertussis* and it can be very serious or even fatal in young children and babies. In the UK, since 1950, young children and babies have been protected from Whooping Cough by vaccination and this reached the target of 95% of children by 1970. By this time there were very few cases.



Bordetella pertussis

In 1974, a medical journal published a report stating that the vaccine had damaged the nervous system in some children. Newspapers spread alarming information to parents throughout the UK and the percentage of children vaccinated fell to 30%.

**“Whooping Cough vaccine can
cause brain damage and death!”**

In 1978, the number of cases of Whooping Cough had risen to 2000 and by 2012 it was 7 730. The antibiotic drugs used to treat Whooping Cough were becoming ineffective.

The negative media reports were discredited and by 2017 the percentage of young children vaccinated in the UK had gone back up to 73%. The target remains at 95%.

Use the information to answer the questions.

- (a) (i) State the scientific name of the micro-organism which causes Whooping Cough. [1]

.....

- (ii) Complete the sentence, by choosing words from the list: [2]

antibodies antigens lymphocytes phagocytes

Micro-organisms which cause infections carry

The in the bloodstream then produce large numbers of

..... which can destroy the infecting micro-organisms.



(b) Choose the correct answer to complete the sentence. [1]

The percentage of children vaccinated against Whooping Cough decreased after 1974 because of newspaper articles which said

- A The vaccine had become less effective.
- B The vaccine had serious side effects.
- C The disease had become less serious.
- D Cases of the disease were reducing.

Answer

(c) (i) Calculate the percentage increase in cases of Whooping Cough between 1978 and 2012. [2]

Use the formula:

$$\frac{\text{number of cases in 2012} - \text{number of cases in 1978}}{\text{number of cases in 1978}} \times 100$$

Percentage increase =

(ii) State the evidence that the increase in the number of cases of Whooping Cough was linked to a reduction in the percentage of children being vaccinated. [2]

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Examiner
only

- (d) (i) By 2017, the number of cases of Whooping Cough had reduced. The NHS continued to encourage the vaccination of young children and pregnant women.

Use the information on page 11 to give **two** reasons why it was necessary to continue to encourage these vaccinations. [2]

.....

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.....

- (ii) In some countries the vaccination of children is compulsory. Suggest why some people object to compulsory vaccination. [1]

.....

.....

- (e) The following six sentences explain why antibiotics became ineffective for treating Whooping Cough, but they are not in the correct order.

- 1 Many doctors over-used the antibiotics.
- 2 The resistant bacteria multiplied rapidly.
- 3 The antibiotics could no longer be used to kill the bacteria.
- 4 Almost all the bacteria became resistant to the antibiotics.
- 5 Doctors discovered antibiotics which killed Whooping Cough bacteria.
- 6 Some bacteria survived because they were resistant to the antibiotics.

Complete the list below to put the six sentences in the correct order by writing the missing numbers on the dotted lines. [2]

5

.....

6

.....

4

.....

13



5. Himalayan balsam (*Impatiens glandulifera*) is a non-native, invasive species found in the UK. Himalayan balsam grows rapidly, out-competing many local species.

Himalayan balsam in flower



In laboratory trials, scientists found that a fungus could kill Himalayan balsam.



Leaf of Himalayan
balsam infected
with fungus

Scientists investigated the possibility of using fungus to reduce the growth of the invasive plant on a larger scale.

- (a) State the term that describes the use of a living organism to reduce the growth of an invasive species. [1]

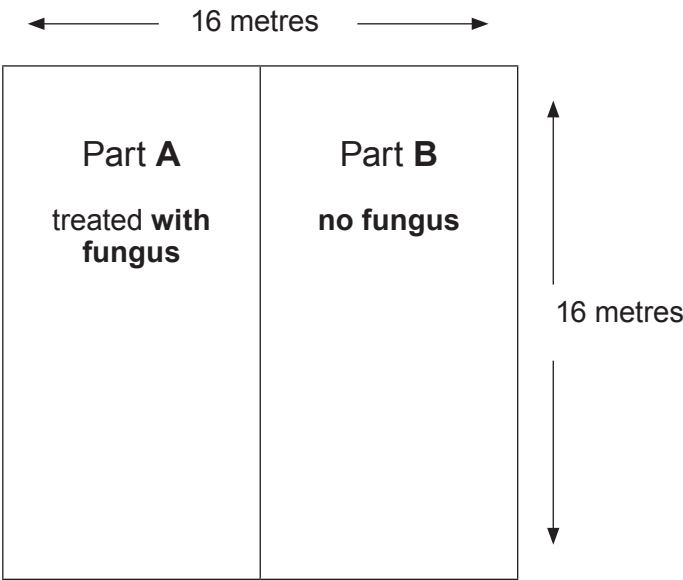
.....



- (b) The scientists marked out an area, 16 metres $\times$ 16 metres, on which many Himalayan balsam plants were growing and were evenly distributed.

They then applied the fungus to half of the section (part **A**). They did not apply any treatment to the other half (part **B**).

Plan of field trial



After six weeks, the scientists estimated the total number of Himalayan balsam plants by quadrat sampling in parts **A** and **B**.

- (i) State why it was important that the quadrats were placed at random. [1]

.....

.....

- (ii) Describe a sampling technique they would carry out, using 1 m² quadrats. [3]

.....

.....

.....

.....

.....

.....



(c) For part **A**, the total number of Himalayan balsam plants found in 12 quadrats was 36.

Calculate

(i) the mean number of plants per 1 m² quadrat in part **A**. [1]

..... plants / m²

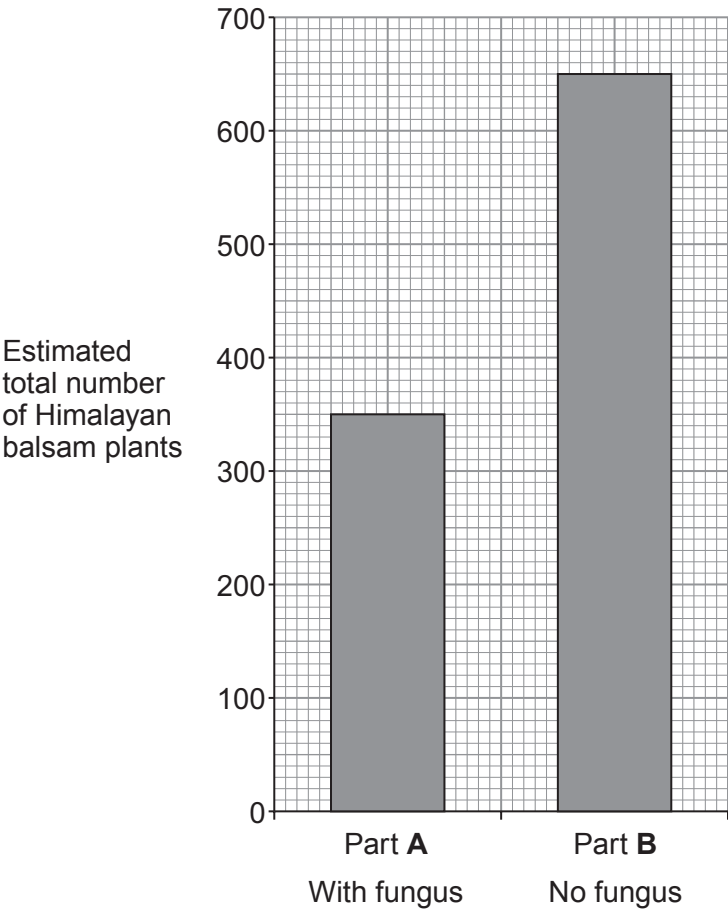
(ii) I. the area of part **A**. [1]

..... m²

II. the estimated total number of plants in part **A**. [1]

..... plants

(d) The scientists carried out their investigation a further 5 times to be confident that their results were repeatable. The mean results of all their investigations are shown in the bar chart.



Examiner
only

(i) State the conclusion that could be drawn from these results. [1]

.....

.....

(ii) State how the scientists could test that the results were reproducible. [1]

.....

.....

(e) (i) The scientists went on to investigate the effects of the fungus on many native plant species. State why this was necessary. [1]

.....

.....

(ii) State how biodiversity would be affected if the Himalayan balsam plants were not controlled. [1]

.....

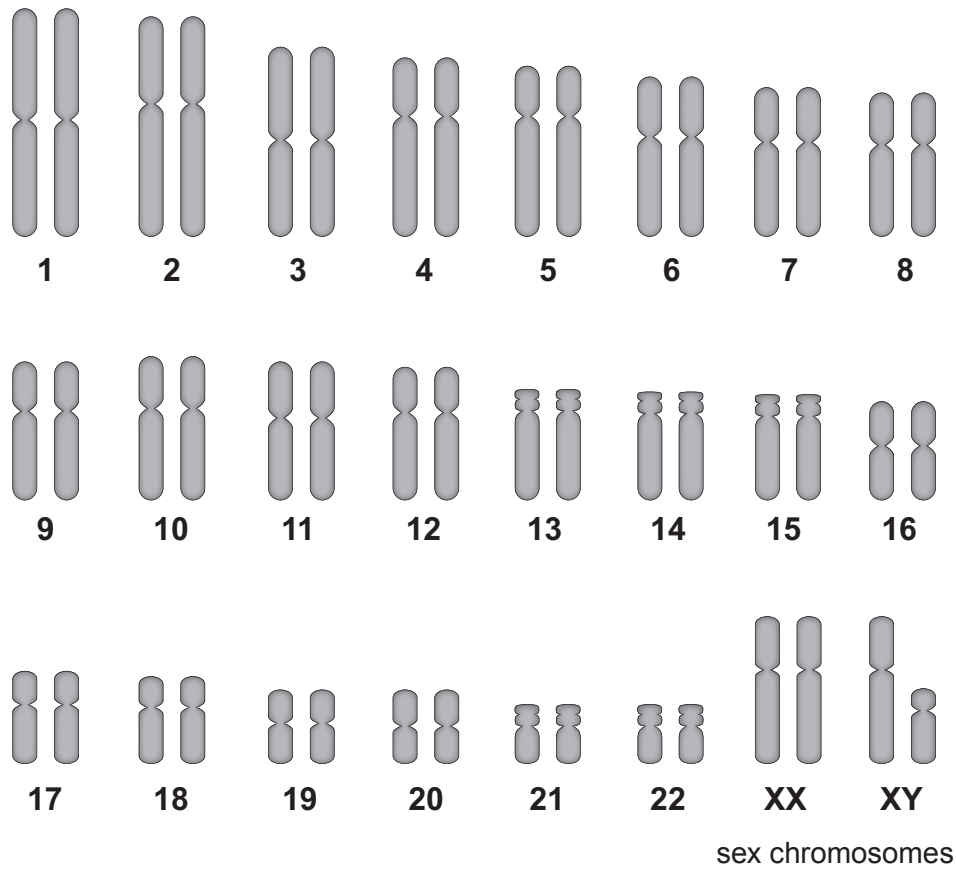
.....

12



Examiner
only

6. (a) The image shows human chromosomes.



- (i) State the number of pairs of chromosomes present in a single body cell of a human. [1]
.....
- (ii) State which sex chromosomes are present: [1]
in body cells of a man;
in body cells of a woman.



(iii) Complete the Punnett square below to show the sex chromosomes in the gametes of a male and female parent and in their possible offspring. [2]

		Male Parent	
Female Parent	Gametes		

(b) Height in humans is controlled by many genes as well as by environmental factors. Females usually reach their full height by age 18 and males by 25.

Students investigated the heights of male and female airline cabin crew. All cabin crew must be between 1.58 m and 1.90 m in height and between 18 and 45 years of age.

The students said that they expected the males to be taller than the females.

They collected their data by selecting seven males and seven females at random and asking them to state their heights. They gave their heights in feet and inches and the students converted the data into metres.

The results, expressed to two decimal places, are shown in the table below.

Heights of females (m)	Heights of males (m)
1.80	1.84
1.78	1.80
1.83	1.72
1.68	1.70
1.75	1.61
1.82	1.81
1.69	1.73
mean height =	mean height = 1.74



Examiner
only

(i) **Complete the table** by calculating the mean height for females. [2]
Space for working.

(ii) I. State the hypothesis that the students were testing in their investigation. [1]

.....

.....

II. State whether the results of the investigation support their hypothesis, giving the reason for your answer. [1]

.....

.....

.....

III. Give **one** way in which the strength of the evidence could be improved. [1]

.....

.....

(c) State **one** source of inaccuracy in the method. [1]

.....

.....

10



7. Cystic Fibrosis (CF) is a serious medical condition which affects the lungs. It results from a mutation in the DNA of a single gene which gives rise to a recessive allele.

(a) Explain the meaning of the term *recessive allele*.

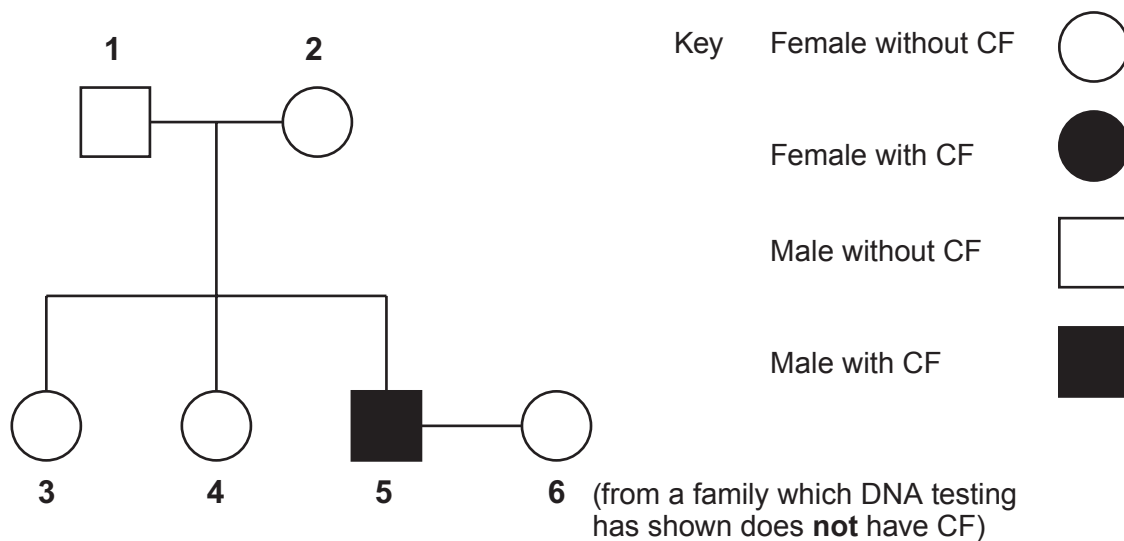
[2]

.....

.....

.....

(b) The diagram below shows part of the family tree of a family which has one member who has CF.



(i) State the effect of a mutation on DNA.

[1]

.....

.....

(ii) From the family tree:

I. State the numbers of **two** individuals who are known to be heterozygous for the allele which causes CF.

[1]

.....

II. Suggest what advice a genetic counsellor would give individuals **5** and **6** about the chance of them having a child with CF. Explain your answer.

[2]

.....

.....

.....

.....

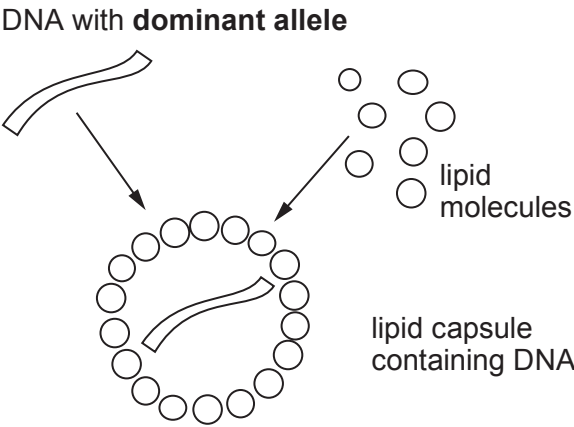


Examiner
only

- (c) CF is usually treated by intensive physiotherapy which can be very stressful. Medical scientists are now working to develop an alternative treatment called *gene therapy*. This process is summarised in the flow chart below.



Physiotherapy for CF



Gene therapy for CF

- (i) Lipid capsules are introduced into the lungs. State how this would be done. [1]

.....

.....

- (ii) In order for this therapy to be successful two further difficulties must be overcome. Describe these **two** difficulties and suggest why this therapy is **not** a permanent cure. [3]

.....

.....

.....

.....

.....

.....

.....

END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE – CONTINGENCY

3400U40-1



WEDNESDAY, 22 JUNE 2022 – AFTERNOON

BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

FOUNDATION TIER

1 hour 45 minutes

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 7(b)(ii) is a quality of extended response (QER) question where your writing skills will be assessed.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	9	
3.	9	
4.	7	
5.	9	
6.	7	
7.	12	
8.	10	
9.	10	
Total	80	



JUN223400U40101

Answer **all** questions.

1. (a) The following statements relate to micro-organisms and disease. [2]
Using a ruler, draw lines to join each term with its correct meaning.

Term		Meaning
Lymphocyte	•	• A micro-organism which causes disease
Pathogen	•	• A medicine that kills bacteria
Antibiotic	•	• A type of white blood cell
Antigen	•	• A molecule recognised by the immune system

- (b) State **two** ways that micro-organisms can be spread between people. [2]

.....
.....

- (c) (i) State the type of micro-organism that causes Chlamydia. [1]

.....

- (ii) The common cold is caused by a virus. State why antibiotics cannot be used to treat the common cold. [1]

.....

- (iii) Underline **one** reason why doctors are now prescribing fewer antibiotics. [1]

Bacteria are becoming immune to the antibiotics.

Bacteria are becoming resistant to the antibiotics.

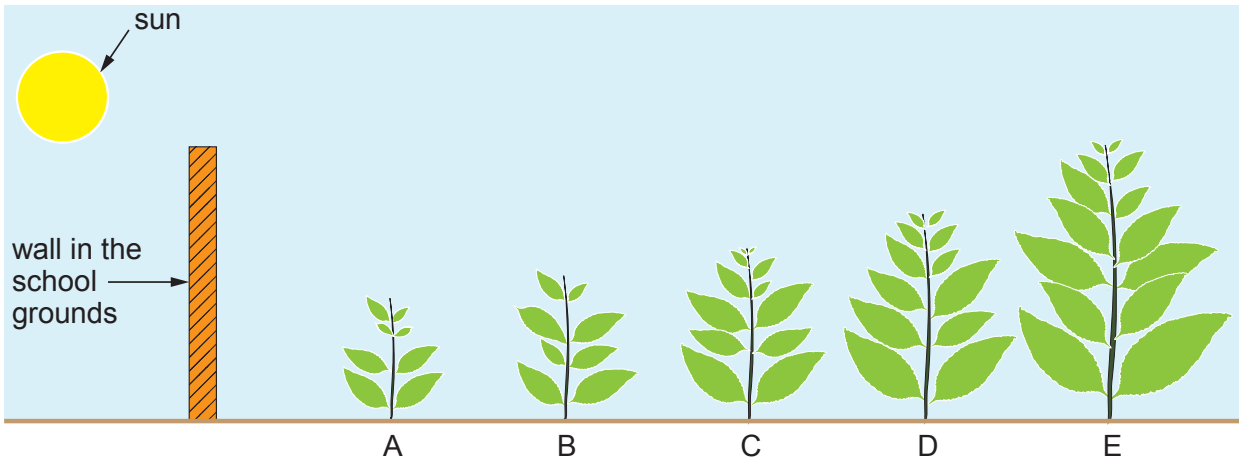
Humans are becoming immune to the antibiotics.

Humans are becoming resistant to the antibiotics.



2. Amber and Mark carried out an experiment. They investigated how the distance from a wall in the school grounds affected the height of plants. They grew five plants at different distances from the wall. This is shown in **Image 2.1**.

Image 2.1



They measured the height of the plants after 1 year.
The results are shown in **Table 2.2**.

Table 2.2

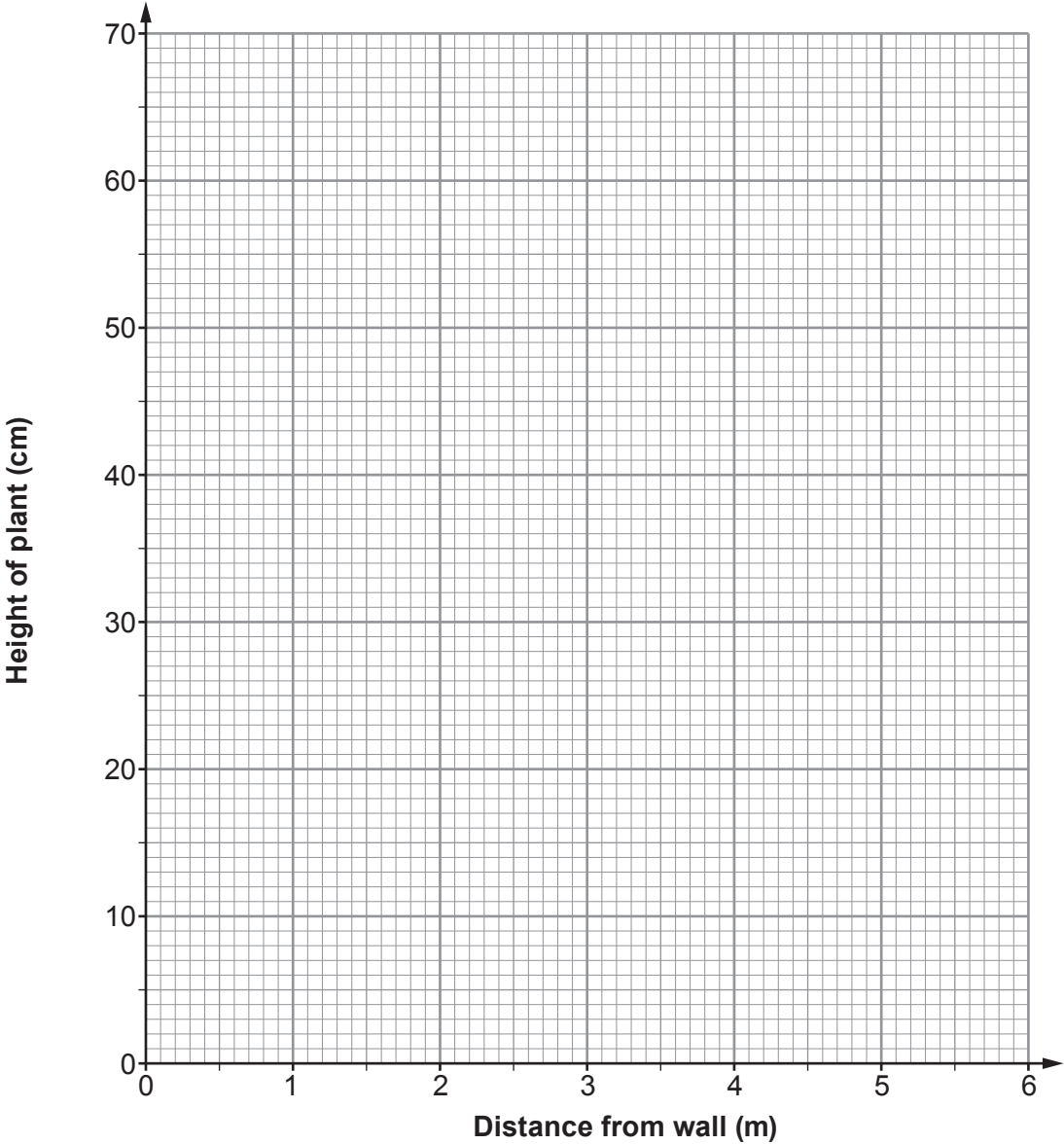
Plant	Distance from wall (m)	Height of plant (cm)
A	1	20
B	2	33
C	3	39
D	4	54
E	5	64

- (a) Use the results in **Table 2.2** to draw a line graph on **Graph 2.3** by: [3]
- (i) plotting the points;
 - (ii) joining the plots using a **ruler**.



Examiner
only

Graph 2.3



(b) Describe the trend shown in the results. [1]

.....

.....

(c) Suggest **two** environmental factors which could have affected the height of the plants. [2]

.....



Examiner
only

(d) Suggest **one** way in which Amber and Mark could improve their confidence in the results. [1]

(e) Amber and Mark did another experiment in a laboratory. They gave five plants different carbon dioxide concentrations. They measured the heights of the plants after one year.

Identify the following for this experiment:

(i) Independent variable [1]

(ii) Dependent variable [1]

9



3. Malaria kills a large number of people in the world each year. It is caused by the single celled organism called *Plasmodium*, which is spread by female mosquitoes (*Anopheles gambiae*).



Female mosquito (*Anopheles gambiae*)

In 1898, some Italian scientists developed a method to control malaria which they called bonifica. They used this approach in a small test area in Italy.

Bonifica involved:

- spraying insecticides in people’s houses;
- hanging mosquito nets in windows of houses;
- pouring petrol on the water in the marshes, where mosquitoes reproduce.

There were 99 new cases of malaria in the test area during the year before bonifica. In the year after there were no new cases.

More recently, anti-malarial tablets have been developed along with a vaccine against *Plasmodium* and the use of biological control.

- (a) (i) Use the information above to complete the following table by writing True or False against **each** statement. [3]

Statement	True or False
In 1899 there were no new malaria patients in the test area.	
Insecticides kill <i>Plasmodium</i> .	
There is evidence that bonifica was successful.	
<i>Plasmodium</i> is passed directly from person to person.	

- (ii) State the genus of the mosquito that spreads *Plasmodium*. [1]

.....



Examiner
only

(b) (i) State the purpose of the nets that were put up at the window. [1]

.....

.....

(ii) Suggest why bonifica could be dangerous to the people who live near the marshes where the mosquitoes reproduce. [1]

.....

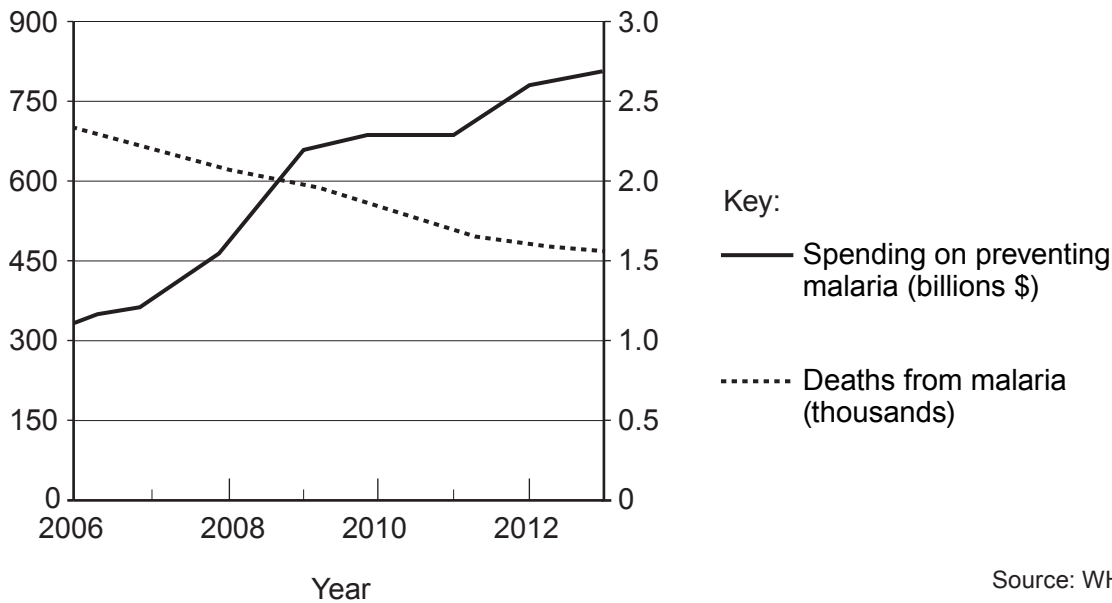
.....

(c) **Graph 3.1** shows the number of annual global deaths from malaria and the amount of money spent on preventing malaria from 2006 to 2013.

Graph 3.1

Deaths from malaria
(thousands)

Spending on preventing
malaria (billions \$)



State the relationship between the money spent on preventing malaria and the number of deaths from malaria. [1]

.....

.....

.....



Examiner
only

- (d) Some of the modern methods used to control malaria include the use of vaccines and biological control.
Choose words from the list to complete the following sentences. [2]

starving preventing killing protecting

Vaccines work by a person getting malaria.
Biological control works by the mosquito that is spreading malaria.

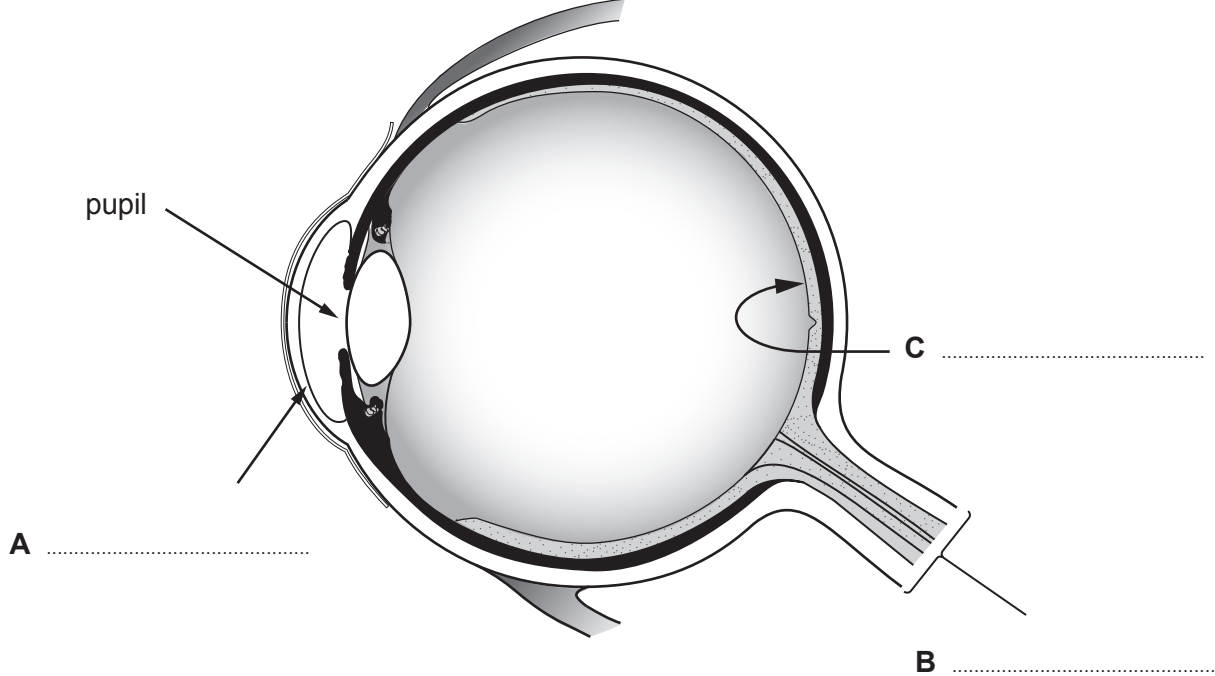
9

3400U401
09



4. **Image 4.1** shows a section through the human eye.

Image 4.1



- (a) (i) Complete the labels **A–C** on **Image 4.1**. [3]
- (ii) State the function of the lens. [1]

.....

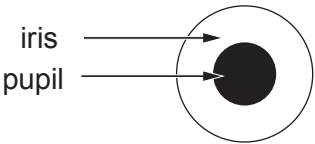
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Examiner
only

- (b) The pupil allows light to enter the eye. The pupil can change in diameter depending on the light intensity. **Image 4.2** shows the appearance of the front of the eye in dim light.

Image 4.2



Use the following information to calculate the area of the pupil. [2]

radius, $r = 4.0 \text{ mm}$

area of pupil = $3.14 \times r^2$

Area of pupil = mm^2

- (c) State what would happen to the area of the pupil if a bright light is directed into the eye. [1]

.....

7

3400U401
11

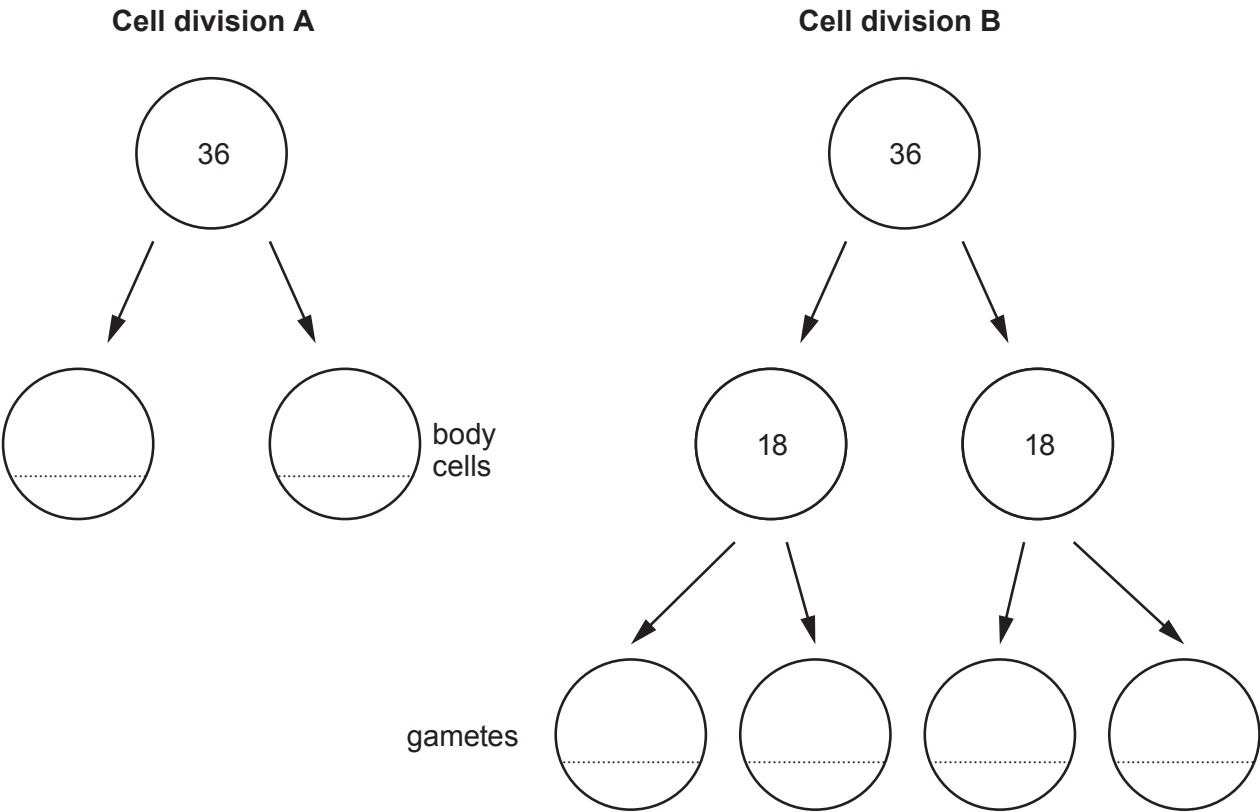


5. Chromosomes are structures made of DNA and are found inside the nucleus of cells. The meerkat (*Suricata suricatta*) has 36 chromosomes in its body cells.



- (a) **Image 5.1** shows two different types of cell division in the **meerkat**.

Image 5.1



- (i) **Complete Image 5.1** by writing in the number of chromosomes in **each** body cell and gamete. [2]
- (ii) State the name of cell division **A**. [1]

.....

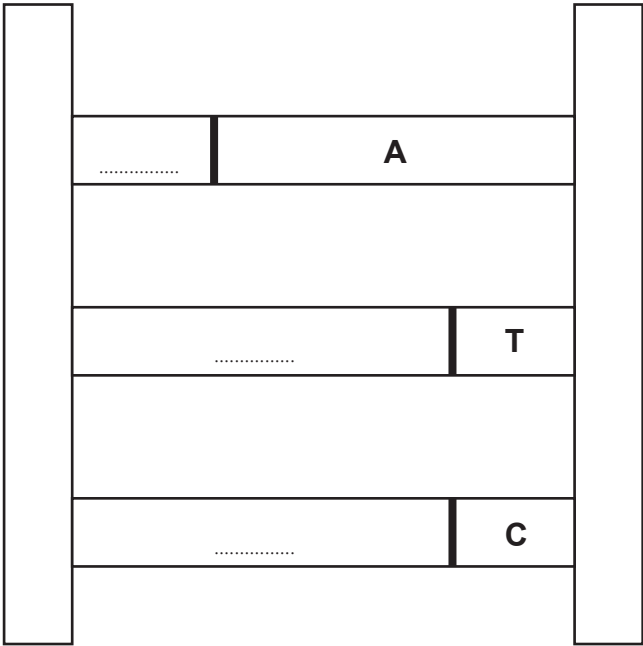


(iii) State the disease that occurs when cell division **A** is uncontrolled. [1]

.....

(b) **Image 5.2** shows part of a DNA molecule. Complete **Image 5.2** by adding the correct letters to represent the names of the missing bases. [2]

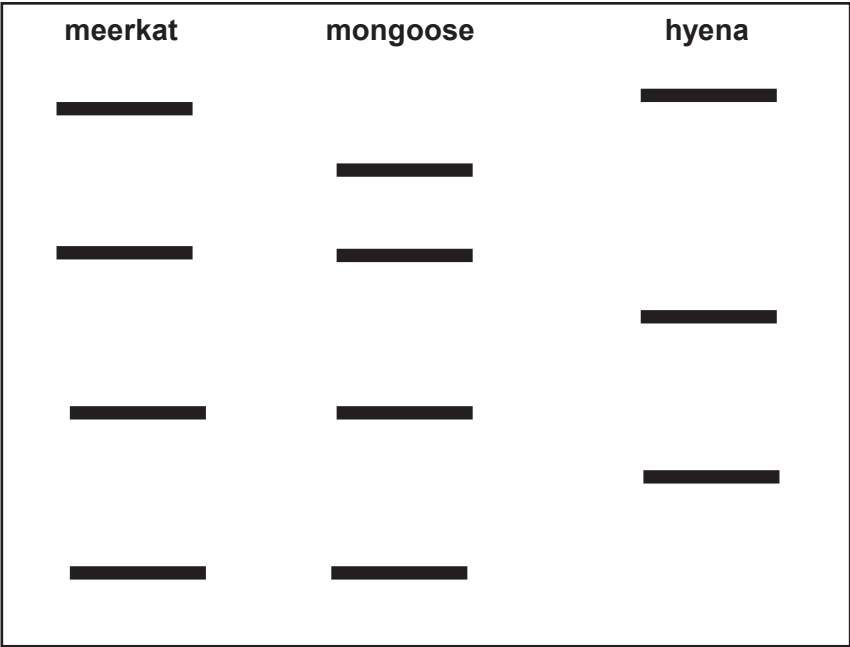
Image 5.2



Examiner
only

- (c) Scientists use DNA profiling to find out how closely related different species of animals are. The DNA profiles of three different animals are shown in **Image 5.3**.

Image 5.3



Use the DNA profiles in **Image 5.3** and your knowledge to answer the following questions:

- (i) State the **two** animals which are most closely related. Explain your answer. [2]

Animals

Explanation

.....

- (ii) State **one** other use for DNA profiling. [1]

.....

9



6. Image 6.1 shows a BATAK machine being used to measure an individual's reactions.

Image 6.1



The method for using the BATAK machine is:

- The lights in the buttons flash randomly, for differing lengths of time.
- The individual has to hit each button when it flashes.
- The number of lights successfully hit in 1 minute is recorded.

Table 6.2 shows the results recorded for five year 11 students. Use the data to answer the questions that follow.

Table 6.2

Name	Number of lights hit successfully in 1 minute			
	Attempt 1	Attempt 2	Attempt 3	Mean
Sean	31	34	39	35
Mark	36	38	41	
Eleri	40	41	42	41
Gwen	29	35	40	35
Gruffydd	21	22	23	22

All the students except Gruffydd did their test during the first lesson of the day. Gruffydd did his test in the last lesson of the day.

- (a) Calculate the mean number of lights hit in 1 minute by **Mark**.
Write your answer to the nearest whole number in Table 6.2.
Space for working

[1]



Examiner
only

(b) State the name of the student with the fastest reactions. [1]

.....

(c) Suggest why Gruffydd's results are not valid in this investigation. [1]

.....

.....

(d) State **two** other variables that should have been controlled in this method. [2]

.....

.....

(e) State **two** properties of reflex actions. [2]

.....

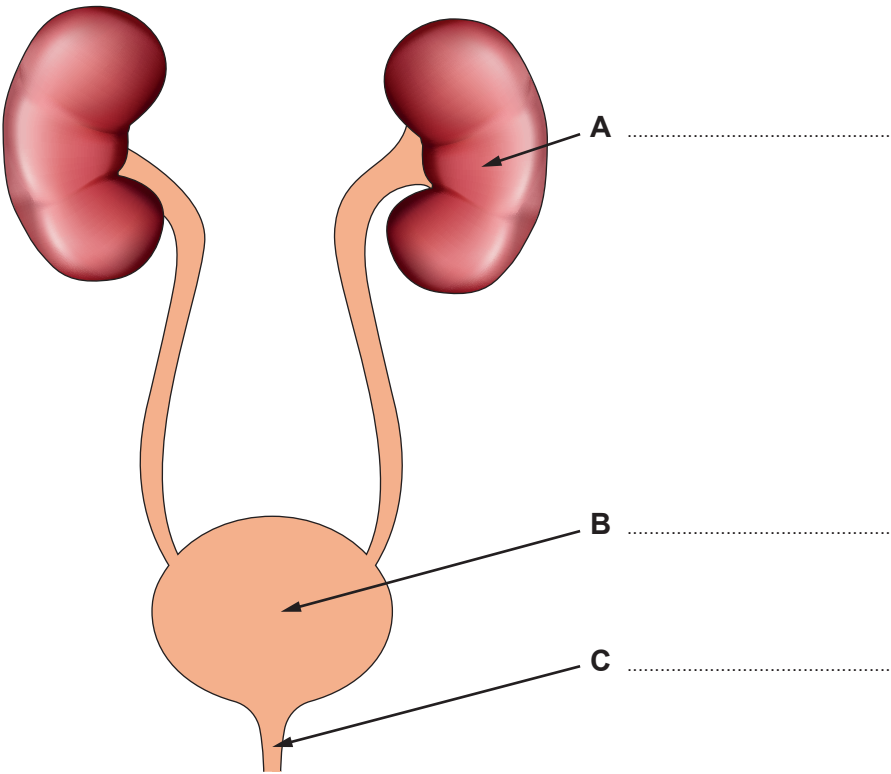
.....

7



7. **Image 7.1** is a diagram of the human excretory system.

Image 7.1



- (a) (i) Label structures **A–C** on **Image 7.1**. [3]
- (ii) **Draw an arrow** on **one** ureter to show the direction that the urine flows. [1]

(b) **Table 7.2** shows the composition of the urine of three people.

Table 7.2

Name	Presence or absence in urine			
	Salt	Water	Urea	Glucose
James	✓	✓	✓	×
Ethan	✓	✓	✓	×
Sophie	✓	✓	✓	✓

✓ indicates present, × indicates absent

- (i) Sophie has a medical condition. Use the information in **Table 7.2** to state the name of the condition. Give a reason for your answer. [2]

.....

.....



Examiner
only

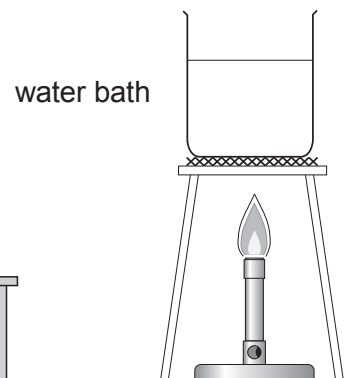
- (ii) Describe how a sample of urine could be tested for the presence of glucose. You must describe how you would use all the apparatus in the diagram and the colour that would be observed for Ethan's and Sophie's samples. [6 QER]



eye protection



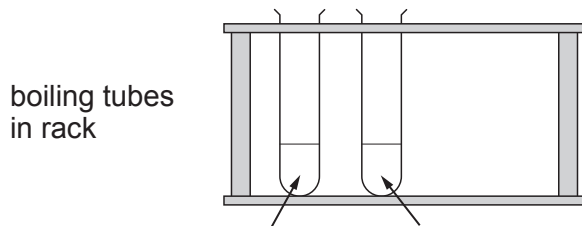
boiling tube holder



water bath



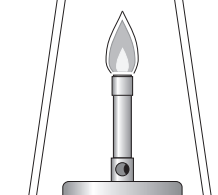
Benedict's reagent



boiling tubes
in rack

Ethan's urine sample

Sophie's urine sample

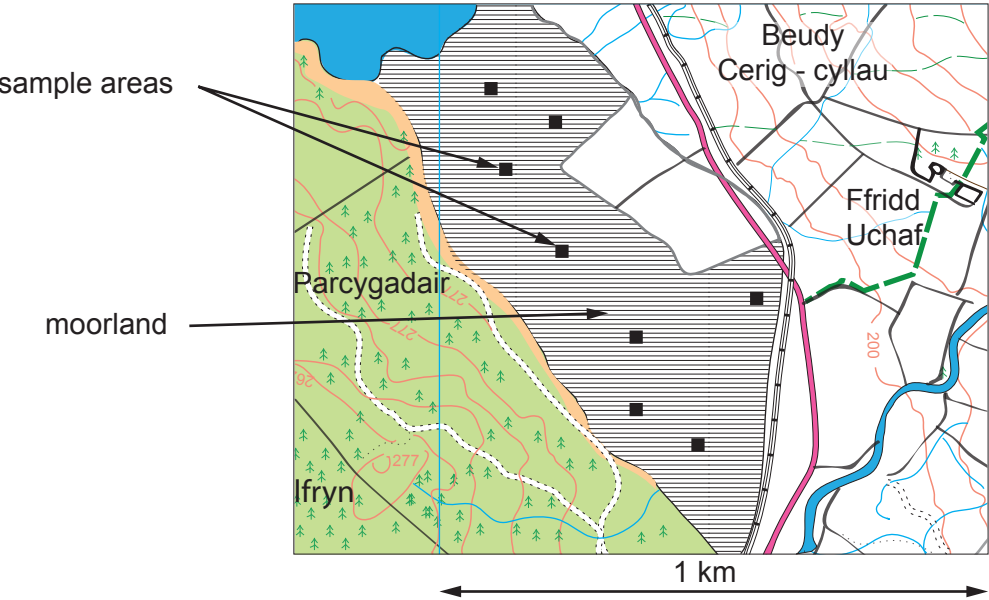


Bunsen burner



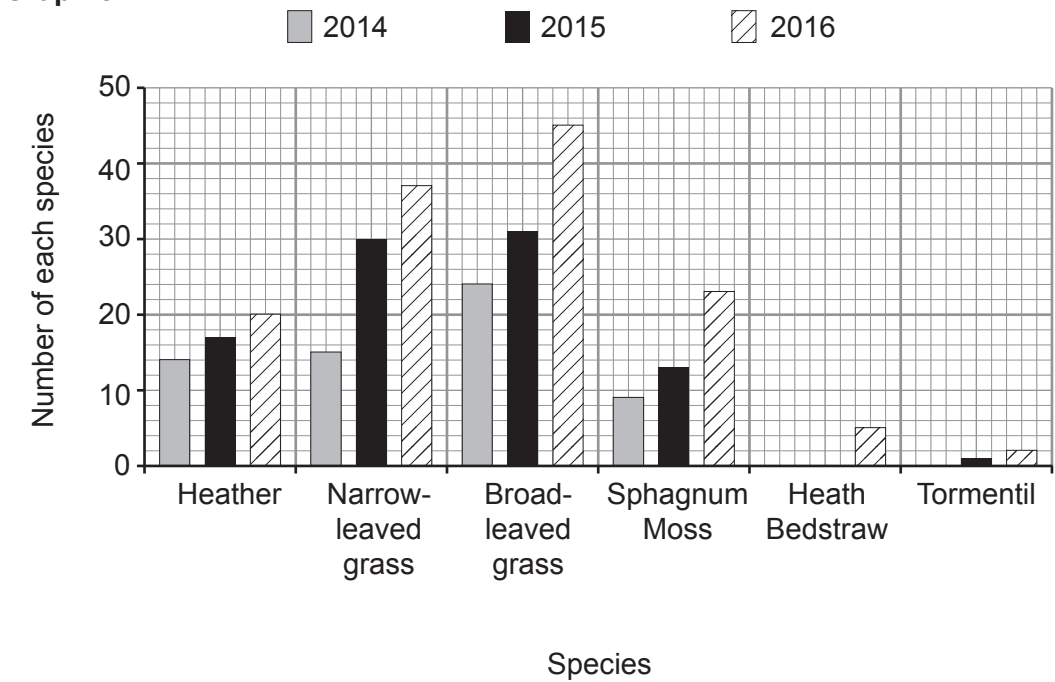
8. A group of students investigated the biodiversity of a moorland over a period of three years. Two years before the investigation started, the drainage ditches in the moorland had been closed in order to increase the water content of the land. **Image 8.1** is a map showing the moorland (shaded area) being investigated. The black squares represent locations of eight sample areas. Each sample area is 625 m² (25 m × 25 m).

Image 8.1



The results of the survey are shown in **Graph 8.2**.

Graph 8.2



Examiner
only

- (a) Describe the method that the students would have used to investigate the abundance of plant species in the moorland. [4]

.....

.....

.....

.....

.....

.....

- (b) (i) Calculate the percentage change in narrow-leaved grass between 2014 and 2016. [2]

Percentage change = %

- (ii) State **two** pieces of evidence that show that the biodiversity of the moorland increased between 2014 and 2016. [2]

.....

.....

.....

.....

- (c) (i) Suggest why the students sampled several areas over the whole moorland. [1]

.....

.....

- (ii) Suggest how, over the three years, the students made sure the investigation was a fair test. [1]

.....

.....

10



9. A peach is a fruit which has tiny hairs on its surface. A nectarine is a type of peach where the hairs are absent. The production of hair on the surface is controlled by a single gene. A dominant allele codes for the production of hair. A mutation to this gene produces a recessive allele which does not produce hair. **Image 9.1** is a picture produced by an electron microscope showing the hair growing on the surface of a peach.

Image 9.1



Peach scab disease is caused by the fungus *Cladosporium carpophilum* and can affect both peaches and nectarines. Fungi reproduce by producing spores, which are spread by wind and rain. When the spore lands on the surface of a fruit, the fungus begins to grow and scabs soon develop. Nectarines are more likely than peaches to develop scab disease. **Image 9.2** is a picture of a nectarine showing scabs caused by *Cladosporium carpophilum*.

Image 9.2

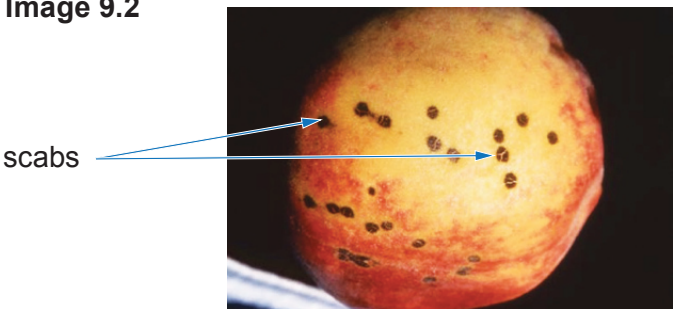
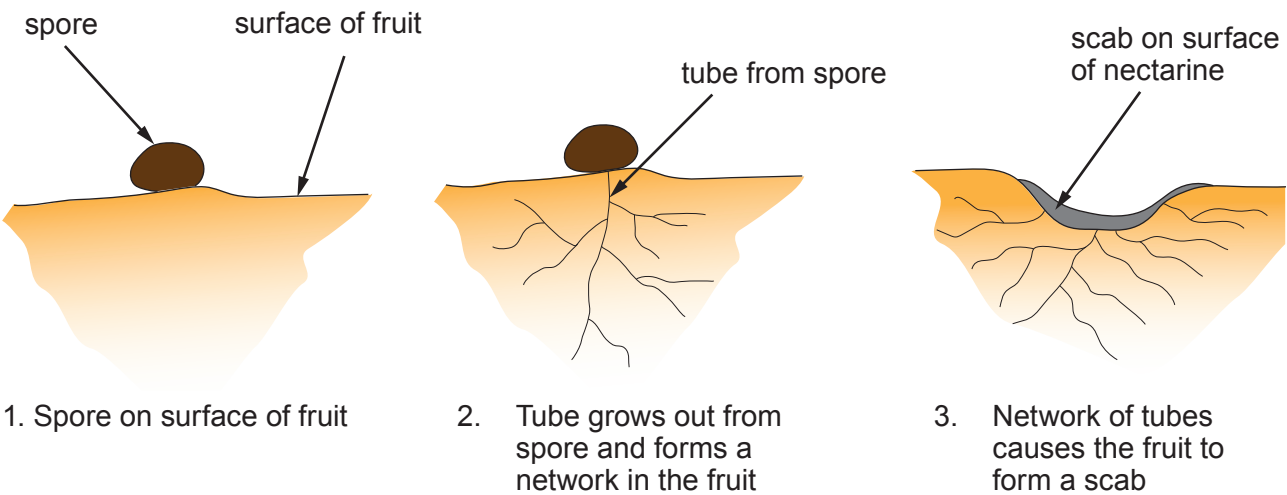


Image 9.3 shows a fungal spore developing and forming a scab.

Image 9.3



Examiner
only

(a) (i) State why scientists use the name *Cladosporium carpophilum* rather than the common name peach scab when discussing this disease. [1]

.....

.....

(ii) Using all the information given, suggest why nectarines are more likely than peaches to develop scab disease. [2]

.....

.....

.....

.....

(b) (i) State what is meant by the term allele. [1]

.....

.....

(ii) State what is meant by a mutation. [1]

.....

.....

(iii) Give **one** example of an **environmental** factor which increases the rate of mutations. [1]

.....

Continued overleaf



Examiner
only

- (c) (i) Use the letters **H** and **h** to show the result of a genetic cross between two peach trees that would result in the production of seeds that could grow into nectarine trees. [3]

Key: **H** = allele for hairy fruit

h = allele for hairless fruit

Phenotype = peach tree × peach tree
Genotype = ×

Gametes		

- (ii) Using your answer from the Punnett square above, state the probability of producing a nectarine tree from this cross. [1]

Probability =

10

END OF PAPER



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3400U20-1



TUESDAY, 17 MAY 2022 – MORNING

BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms

FOUNDATION TIER
1 hour 45 minutes

For Examiner’s use only		
Question	Maximum Mark	Mark Awarded
1.	10	
2.	8	
3.	10	
4.	8	
5.	12	
6.	12	
7.	9	
8.	11	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional pages at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 4(a) is a quality of extended response (QER) question where your writing skills will be assessed.



JUN223400U20101

Answer **all** questions.

1. Cystic fibrosis (CF) is an inherited condition caused by a DNA mutation.

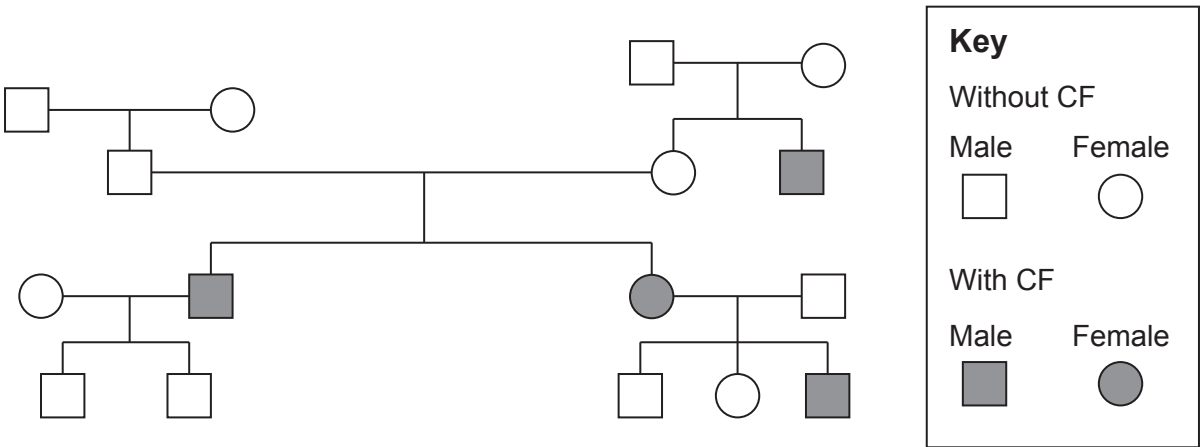
(a) **Complete the sentences** by selecting your answers from the words below. [3]

radiation	increase	random	prevent	regular
-----------	----------	--------	---------	---------

A mutation is a change in DNA.

Ionising can the rate of mutations.

(b) In the family tree below, some people have CF.



(i) Calculate the **percentage of people** in this family tree who have CF. [2]

Percentage = %

(ii) State how many males and females in this family tree have CF. [1]

Males =

Females =



(iii) In the whole population of the UK,

- 0.01% of people have CF
- The ratio of males to females is 1 : 1

From your answers to (i) and (ii), give **two** ways in which this family is different from the whole population of the UK: [2]

I.

II.

(c) CF affects the lungs.

In a treatment, patients with CF are given DNA which does not have the CF mutation.

Complete the following sentences by underlining the correct word. [2]

(i) The treatment is called:

chemotherapy

gene therapy

physiotherapy

(ii) The DNA is given to the patient by:

injection

infusion

inhalation

3400U201
03



2. Images 2.1A and 2.1B show the hazel dormouse (*Muscardinus avellanarius*) in winter and summer.

Image 2.1A

winter



thick fur covering the body and long tail

Image 2.1B

summer



grasping feet
with claws

large eyes for
seeing in dim light

- Dormice are nocturnal (active only at night).
- In summer, they live high up in trees and bushes, eating berries and insects.
- In winter, they hibernate (being completely inactive) and keep warm in nests at ground level.
- Between the years 2000 and 2020 the dormouse population in the UK decreased and it became an endangered species in some areas.
- In 2000, as part of a local action plan, 1 000 dormice were released into an area and their numbers later increased to 1 050.

Use the information to answer the following questions.

(a) State **one** way in which the dormouse **body structure** is adapted to: [3]

(i) survive in low temperatures;

.....

(ii) climb in branches of trees;

.....

(iii) look for food at night.

.....



(b) State **one** way in which the **behaviour** of the dormouse helps it to survive in its environment.

[1]

(c) Using the information, complete the table by writing true or false for each statement. [4]

Statement about the hazel dormouse	True or false
Its habitat is woodland.	true
It eats only plants.	
The species became extinct in the UK in 2020.	
It does not hunt for food in daylight.	
As a result of a local action plan, numbers increased by 5%.	
It belongs to the genus <i>Muscardinus</i> .	

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05



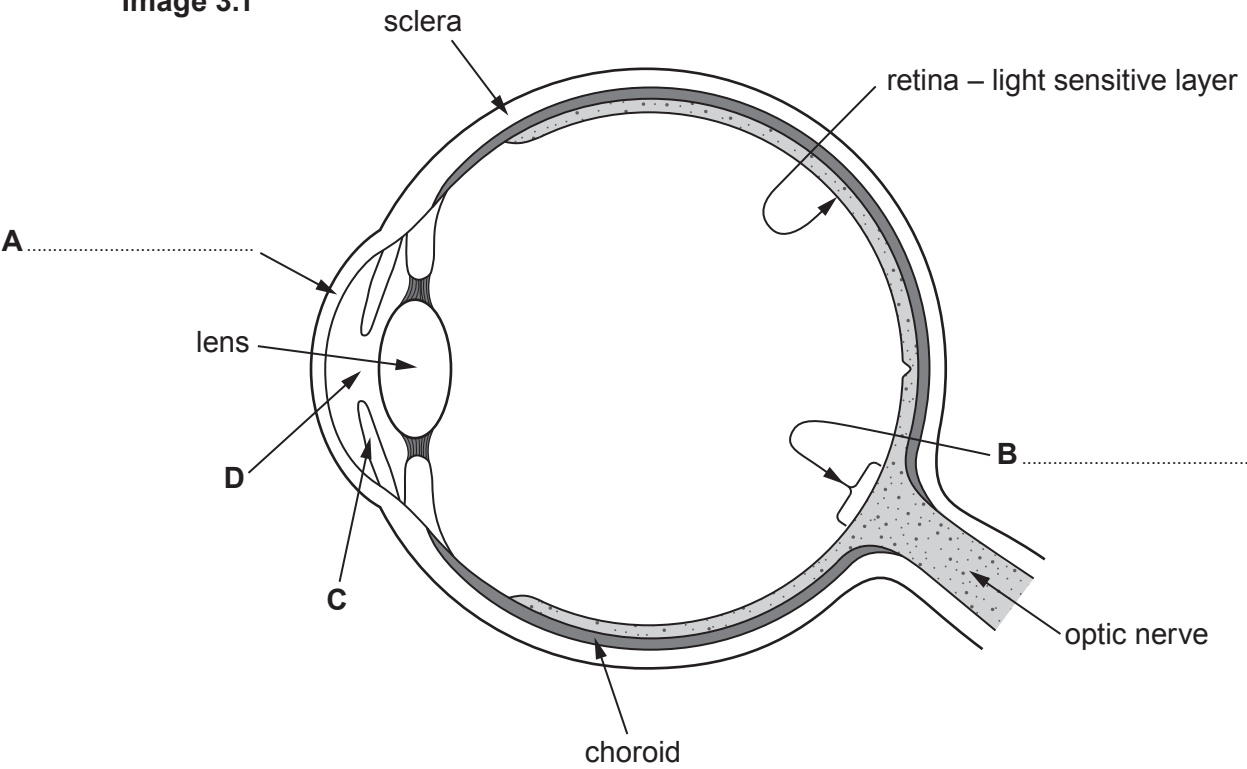
3. (a) **Complete the sentence** by selecting your answers from the words below. [2]

brain	muscles	spinal cord	nerves
-------	---------	-------------	--------

The central nervous system consists of the and the

(b) **Image 3.1** shows a section through the human eye.

Image 3.1



(i) **Label parts A and B on Image 3.1.** [2]



- (ii) Using some labels from **Image 3.1**, complete **Table 3.2** by stating the parts of the eye which match the functions. [3]

Table 3.2

Part of the eye	Function
.....	changes shape to focus light
.....	prevents reflection of light
.....	carries nerve impulses to the brain

- (iii) Joanna leaves a dark room and goes out into bright sunlight.



Explain how parts **C** and **D** in **Image 3.1** change in order to control how much light enters her eyes when she goes into bright sunlight. [3]

.....

.....

.....

.....



4. Serious kidney failure can be treated using a dialysis machine or by a transplant from a donor. **Table 4.1** shows a fact file about the treatment of kidney failure.

Table 4.1

Fact file – Treatment of Kidney Failure		
	Dialysis	Transplant
Percentage of patients surviving after five years	35	97
Waiting time for treatment	2–3 weeks	3–4 years
Usual time spent in hospital	3 days every week for life	one 5-day stay for an operation
Procedure	needles inserted into blood vessels	major surgery
Drugs	anti-rejection drugs not required	anti-rejection drugs needed for life
Diet	special low-salt foods and restricted fluid intake	normal balanced diet and normal fluid intake
Employment and sports	very limited choice	most types of jobs and many sports can be done



Examiner
only

- (a) Using the information in **Table 4.1** and your own knowledge:
Describe the **advantages** of treating kidney failure by a transplant.
Describe the **advantages** of treating kidney failure by dialysis.
Suggest how more people could be encouraged to become donors and explain why this is necessary. [6 QER]

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- (b) Explain why the tissue type of the patient and the donor kidney must be tested before a transplant operation is carried out. [2]

.....

.....

.....

8

3400U201
09



5. (a) **Complete the following** description of Type 2 diabetes by filling in the missing words. [3]

In Type 2 diabetes, body cells do not respond to the hormone
which is produced by the and so the level of
..... in the blood becomes too high.

(b) Researchers in Spain stated the following hypothesis.

‘Drinking coffee reduces the risk of having Type 2 diabetes, high blood pressure and obesity.’

- Scientists working for a large chain of coffee shops carried out an investigation to test this hypothesis.
- They used 2000 volunteers, 1 000 of whom drank coffee every day and the other 1 000 who never drank coffee.
- They recorded the number of volunteers from each group who had Type 2 diabetes, had high blood pressure or were obese.

The results of the investigation are shown in **Table 5.1**.

Table 5.1

Condition	Number of volunteers with the condition	
	Coffee drinkers	Non-coffee drinkers
Type 2 diabetes	100	100
High blood pressure	280	420
Obesity	340	460

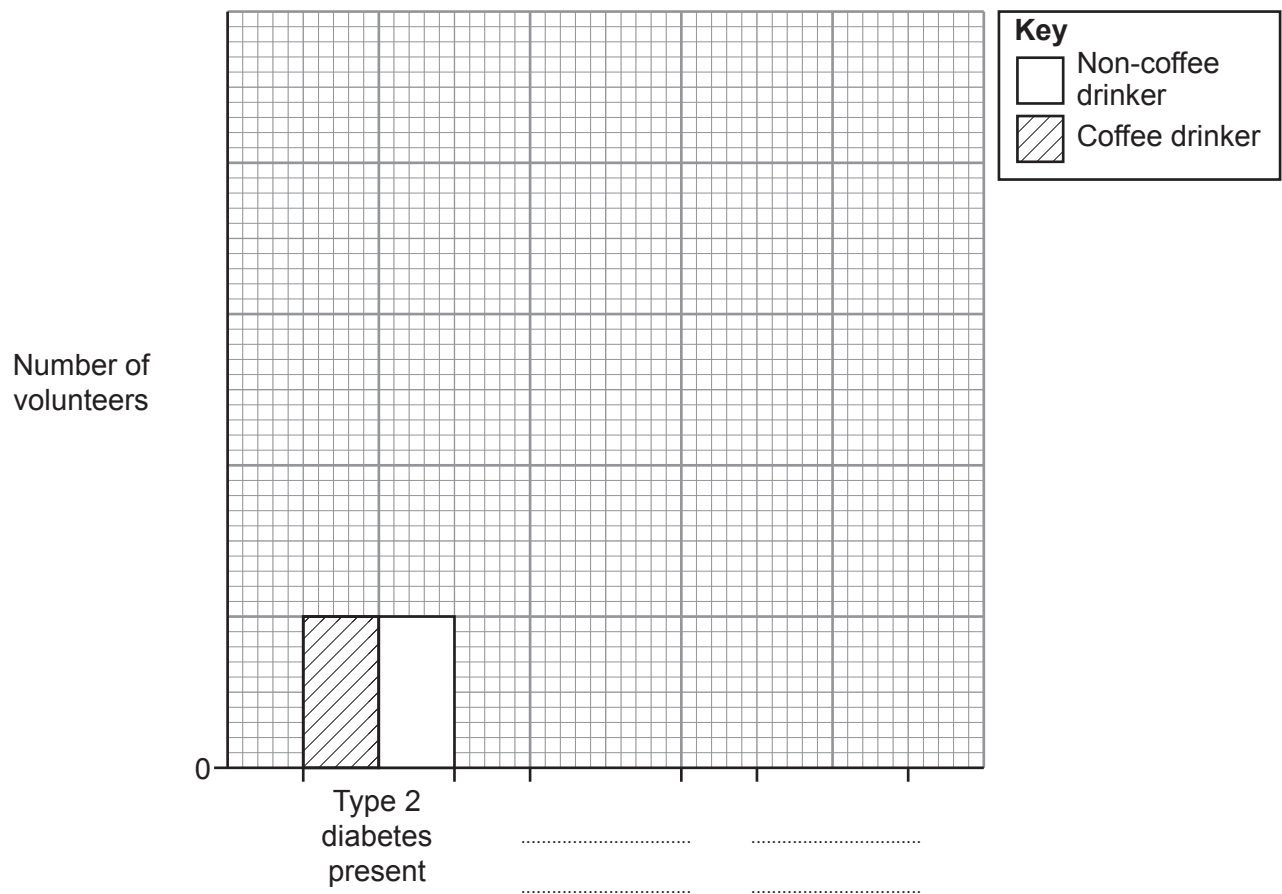


- (i) Complete the bar chart in **Graph 5.2** by:
- [4]
- I. adding a scale for numbers of volunteers.

II. drawing bars of the results for high blood pressure and obesity.

III. labelling your bars.

Graph 5.2



- (ii) The scientists concluded that the hypothesis was only partly supported. Give the reasons for this conclusion.
- [2]

.....

.....

.....

.....



Examiner
only

(iii) State **one** way in which the scientists should have ensured that the investigation was a fair test. [1]

.....

.....

(iv) A doctor said that she did not have confidence in the results because the investigation was biased. [1]

Give **one** reason to support this point of view.

.....

.....

(c) In 2019 the cost to NHS Wales of medical treatments for Type 2 diabetes was £500 million.

Suggest **one** lifestyle change which individuals can make to reduce the risk of developing Type 2 diabetes. [1]

.....

12



6. Limpets (*Patella vulgata*) are animals without backbones which live on rocky seashores, feeding on plants.



Photograph of limpets

- (a) State the scientific term for animals which do not have backbones.

[1]

.....

- (b) Students investigated the density of limpets (number per m^2) on two rocky shores in Anglesey. One shore was sheltered and one was exposed to heavy wave action.

seaweed



A sheltered shore



bare
rocks

An exposed shore

Students' method:

- Select a section on each shore of 300 m^2 .
- Place quadrats, each of area 1 m^2 , at 10 random co-ordinates in each of the two shore sections and collect data.
- Compare the data for the two shores.



Examiner
only

(i) **Table 6.1** shows one part of the students' risk assessment for the investigation.

Complete Table 6.1. [1]

Table 6.1

Hazard	Risk	Control measure
Sharp edges on rocks		

(ii) Describe in detail the techniques the students should use to **place their quadrats at random** and **collect data**. [3]

.....

.....

.....

.....

.....



The results of the investigation are shown in **Tables 6.2 and 6.3.**

Table 6.2

Number of limpets on exposed shore:

Quadrat number	1	2	3	4	5	6	7	8	9	10	Mean number per m ²	Estimated total number in the 300 m ² section of shore
Number of limpets	26	21	22	18	5	21	17	23	19	26	19.8	5940

Table 6.3

Number of limpets on sheltered shore:

Quadrat number	1	2	3	4	5	6	7	8	9	10	Mean number per m ²	Estimated total number in the 300 m ² section of shore
Number of limpets	30	22	26	31	28	25	23	19	31	26		

(iii) **Complete Table 6.3 for the sheltered shore** by calculating: [3]

- I. The mean number of limpets per m².
- II. The estimated total number of limpets in the 300 m² section of the shore.

Space for working



Examiner
only

(iv) I. From these results, state what the students could conclude about the density of limpets when they compared the two shores. [1]

.....

.....

II. Suggest an explanation for this observation. [1]

.....

.....

(v) State which of the quadrats (1 – 10) from the exposed shore shown in **Table 6.2** had an anomalous result and describe what should have been done to take account of this. [2]

.....

.....

.....

12



7. Biological control is used to reduce the numbers of a pest population through the introduction of another species. It has been used with varying success since the 19th century.
- (a) (i) State **two** advantages of this method of control. [2]
- Advantage 1
-
-
- Advantage 2
-
-
- (ii) State **two** disadvantages of this method of control. [2]
- Disadvantage 1
-
-
- Disadvantage 2
-
-
- (b) The whitefly (*Trialeurodes vaporariorum*) is a pest which damages greenhouse crops such as tomatoes. Whitefly numbers can be reduced by using the biological control agent *Encarsia formosa*.

Encarsia formosa fact file

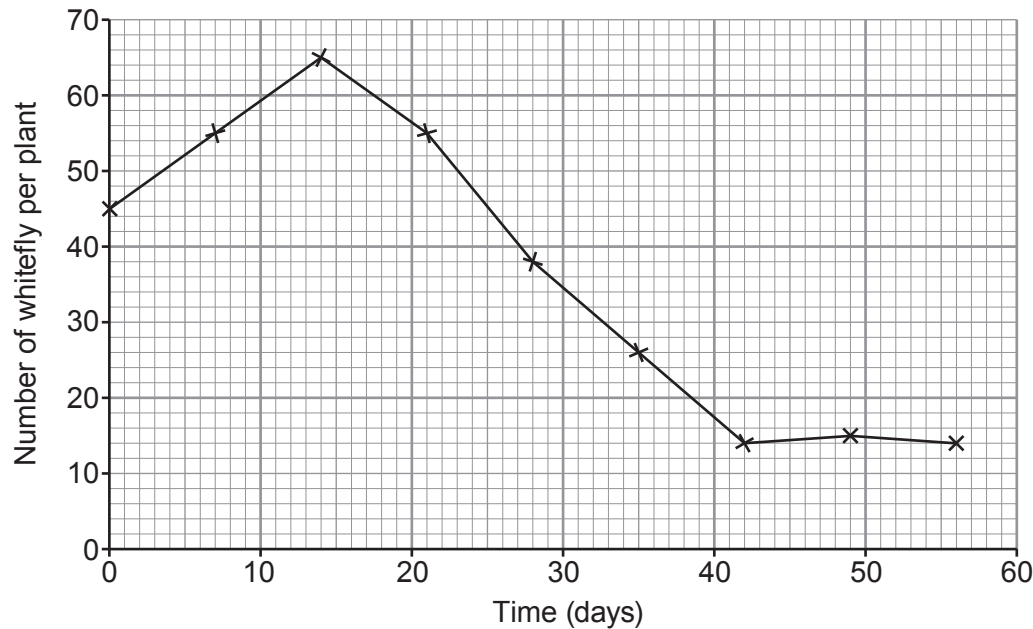
- *E. formosa* is a tiny wasp that lays eggs inside developing whitefly.
- When the eggs hatch, the young wasps kill the developing whitefly from the inside.
- Optimal conditions for *E. formosa* are temperatures over 20°C.
- When daytime temperatures are less than 17°C, *E. formosa* activity is significantly reduced, making it less effective.



Examiner
only

Graph 7.1 shows the number of whiteflies in a greenhouse containing tomato plants. *E. formosa* were introduced on day 7.

Graph 7.1



- (i) I. The use of *E. formosa* to reduce the number of whiteflies is considered to be successful when there are 20 or fewer whiteflies per plant. Use **Graph 7.1** to determine how many days it took for the number of whiteflies to fall to 20 following the introduction of *E. formosa*. [2]

..... days

- II. Suggest a reason why it took this long for the number to fall to 20. [1]

.....
.....

- (ii) Suggest **one** reason why this method of pest control would not be effective to use if whiteflies damaged wheat crops grown in Wales. [1]

.....
.....
.....

- (iii) An alternative approach to reducing pest numbers is to use pesticide. State why it is not appropriate to use pesticide along with *E. formosa*. [1]

.....
.....



8. Hair length in cats is controlled by a pair of alleles. The allele for short hair (**H**) is dominant to the allele for long hair (**h**).



(a) State what is meant by the terms:

(i) allele; [1]

.....

.....

(ii) dominant; [1]

.....

.....

(iii) recessive. [1]

.....

.....



Examiner
only

- (b) (i) A cat breeder crossed a homozygous short-haired cat with a long-haired cat. **Complete the Punnett square** to show the predicted genotypes of the offspring. Use the letters **H** and **h** for the alleles. [2]

Gametes		

- (ii) State the **phenotype** of the offspring in the F1 generation. [1]

.....

- (iii) **Complete the Punnett square** to show the possible genotypes of the offspring if two of the F1 offspring were crossed. [2]

Gametes		

- (iv) Using the results from (b)(iii), state how many kittens would be predicted to be short-haired in a litter of 8 kittens. [1]

.....

- (v) The cat breeder wanted to determine whether one of the short-haired cats was homozygous or heterozygous. She decided to breed the short-haired cat with a long-haired cat. Predict the phenotypes of the offspring you would expect if the short-haired cat was:

I. Homozygous [1]

.....

II. Heterozygous [1]

.....

END OF PAPER

11



Surname	Centre Number	Candidate Number
First name(s)		0



GCSE
3400U20-1



TUESDAY, 16 MAY 2023 – MORNING

BIOLOGY – Unit 2:
Variation, Homeostasis and Micro-organisms
FOUNDATION TIER
1 hour 45 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	6	
3.	9	
4.	10	
5.	6	
6.	14	
7.	8	
8.	11	
9.	9	
Total	80	

ADDITIONAL MATERIALS

In addition to this paper you may require a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.
You may use pencil for graphs and diagrams only.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question 7(b) is a quality of extended response (QER) question where your writing skills will be assessed.



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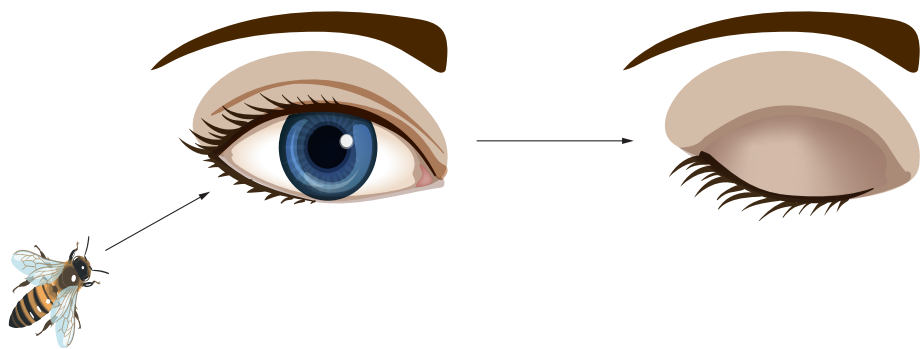
Answer **all** questions.

1. (a) Use some of the words from the box to **complete the sentences** about the nervous system. [4]

cornea	light	chemical	retina	electrical	receptor
--------	-------	----------	--------	------------	----------

Sense organs contain groups of cells which respond to stimuli.
In the eye, cells in the respond to the stimulus of
and send impulses to the brain along neurones.

- (b) The diagram shows how the eye closes automatically (blinks) when something which may be harmful approaches.



Select the scientific term (**A**, **B** or **C**) for this type of action from the list below. [1]

- A** a release action
- B** a relax action
- C** a reflex action

Answer

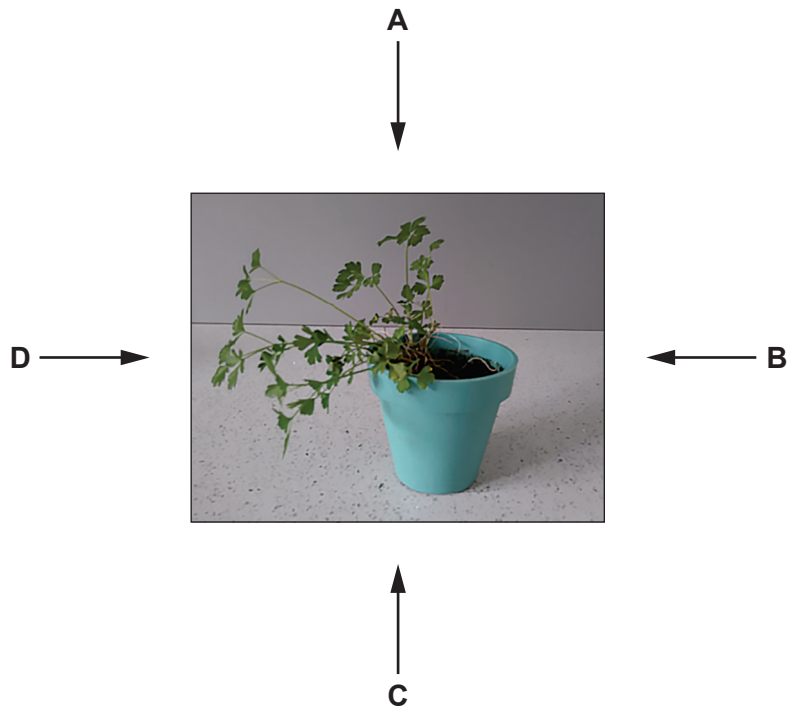
- (c) Apart from happening automatically, state **two** other properties of actions such as blinking. [2]

- 1
- 2



2. **Image 2.1** shows a parsley plant (*Petroselinum sp.*) growing on a windowsill.

Image 2.1



The plant is showing a response to light called positive phototropism.

- (a) (i) State which **one** of arrows **A**, **B**, **C** or **D** on **Image 2.1** shows the main direction of light. [1]

Answer

- (ii) Underline the name of the plant hormone involved in the response. [1]

insulin

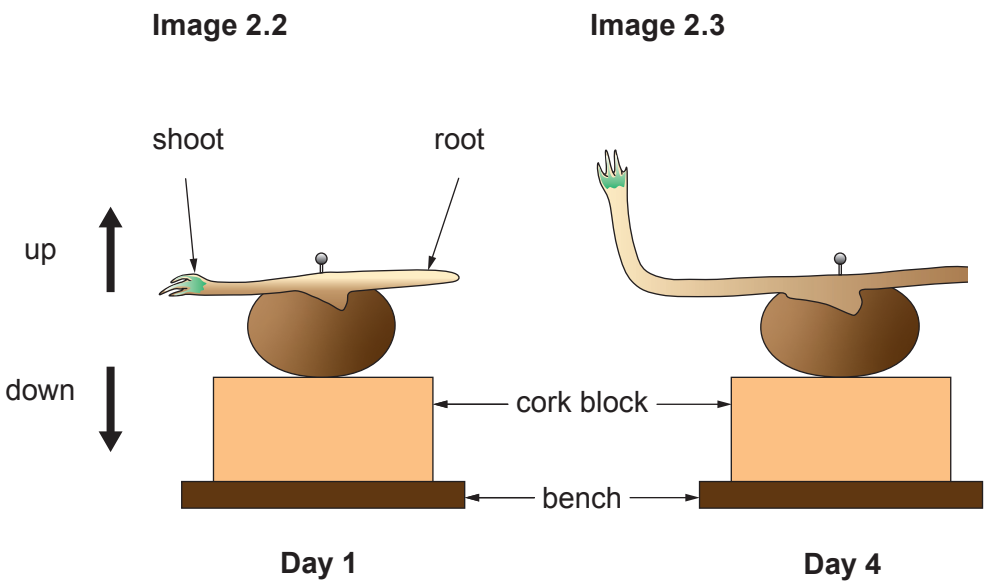
amylase

auxin



(b) **Images 2.2 and 2.3** show an investigation into the effect of gravity on the growth of a seedling which has been placed horizontally on a laboratory bench.

(i) **Complete Image 2.3** by drawing **the root** as you would expect to see it at **day 4**. [2]



(ii) I. State the name of the response of the root to gravity. [1]

II. Explain why this response is important to the seedling. [1]

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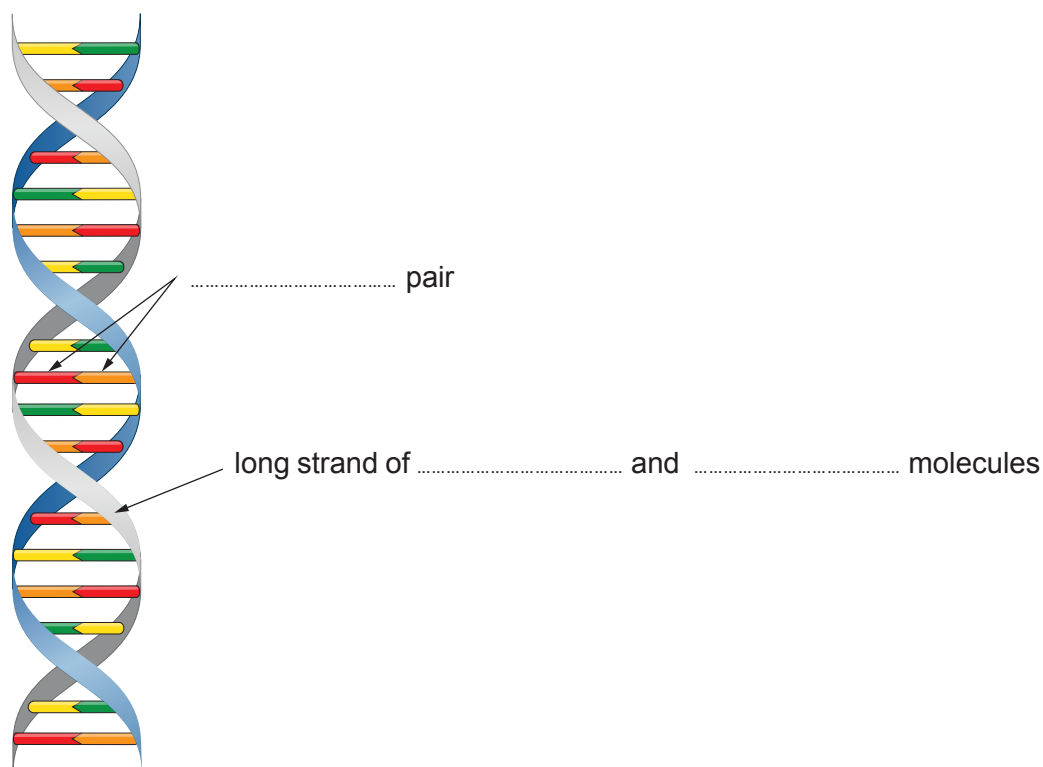


3. (a) **Image 3.1** shows part of a DNA molecule.

(i) **Complete the labels on Image 3.1** by choosing words from the box. [3]

phosphate	salt	base	sugar	amino acid
-----------	------	------	-------	------------

Image 3.1



(ii) The shape of DNA is described as a twisted ladder. State the scientific term for this shape. [1]

.....

(b) The DNA of an individual can be analysed as a genetic profile.

Underline which **one** of the following correctly states how the DNA must first be treated in order to produce the bands in a genetic profile. [1]

Cut it into small pieces.

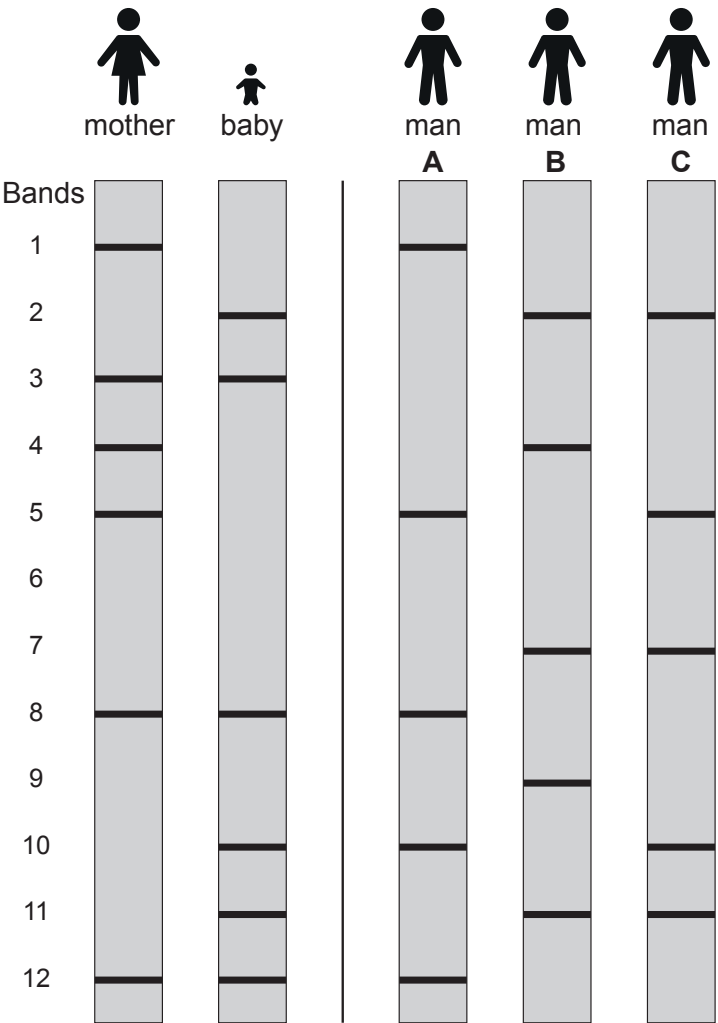
Place it in acid conditions.

Expose it to very low temperatures.



(c) **Image 3.2** shows the DNA profiles of people involved in a paternity test.

Image 3.2



- (i) In this baby's profile, three of the bands have come from the mother and three from the father.

Draw circles around the **three bands** in the **baby's profile** in **Image 3.2** which have come from the **mother**.

[1]

- (ii) Use the DNA profiles to conclude which man (**A**, **B** or **C**) is the baby's father. Explain how you obtained your answer.

[2]

.....

.....

.....

- (iii) Apart from paternity testing state **one** other use for DNA profiling.

[1]

.....



4. In the 1980s young terrapins (*Trachemys scripta*) from North America were sold in pet shops throughout the UK. When the terrapins became adults, many people no longer wanted them and released them into local ponds. The animals can live for up to 50 years.



Young terrapin



Adult terrapin

Table 4.1

Terrapin age	Body length (mm)
3 months	25
2 years	50
12 years	300

After release, adult terrapins eat native animals including invertebrates and vertebrates such as frogs and fish. In some ponds, native species have been reduced or lost completely. Since 1992 a law has banned the import of terrapins into the UK.

Use the information to answer the questions.

- (a) Complete the classification of the terrapin. [3]

Kingdom
Phylum Chordates (Vertebrates)
Class Reptiles
Genus
Species



- (b) (i) Calculate the increase in body length **per year** for an adult terrapin in the **ten years** between 2 years old and 12 years old. [2]

Increase in body length = mm/year

- (ii) Suggest **one** reason why some people did not want to keep the terrapins when they became too big. [1]

.....

.....

- (c) **Complete Table 4.2** by writing true or false for each of the statements. One row has been done for you. [3]

Table 4.2

Statement about terrapins in the UK	True / False
They can live for over 60 years.	False
They do not eat animals with backbones.	
They have reduced the biodiversity in ponds.	
They cannot be brought into the country legally.	
They are an alien species.	

- (d) In the USA, these terrapins are called red-eared sliders and in Welsh they are called terapin clustgoch.

State how scientists from different countries ensure that they are all talking about the same species. [1]

.....

.....



5. The photograph shows a koala (*Phascolarctos cinereus*). These animals live in trees in Australia. Their chromosome number is 16.



(a) (i) **Complete Table 5** about cell division in the koala. [3]

Table 5

	Number
Number of chromosomes in one body cell.	16
Number of cells produced from one cell by mitosis .	
Number of chromosomes in each new cell produced by mitosis .	
Number of chromosomes in each new cell produced by meiosis .	

(ii) State:

I. **one** function of mitosis; [1]

.....

II. the scientific name for the daughter cells produced by meiosis. [1]

.....



- (b) Another Australian animal, the grey kangaroo (*Macropus giganteus*) also has 16 chromosomes but its appearance is very different from the koala.



Underline which **one** of the following statements explains what causes the appearance of these two species to be different. [1]

- They live in different habitats.
- The genes on their chromosomes are different.
- Their diet consists of different types of plants.

6

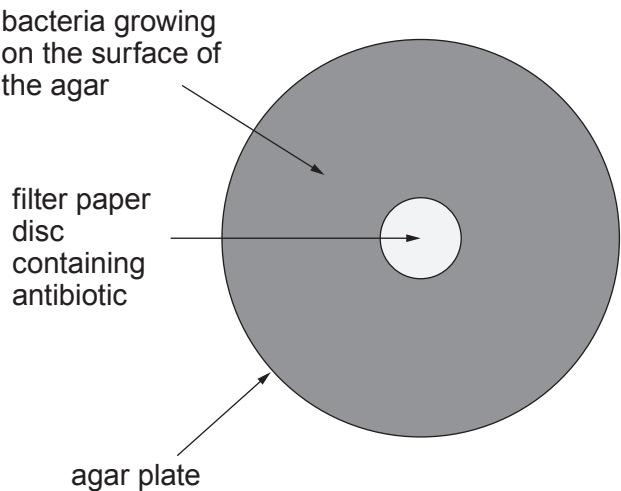
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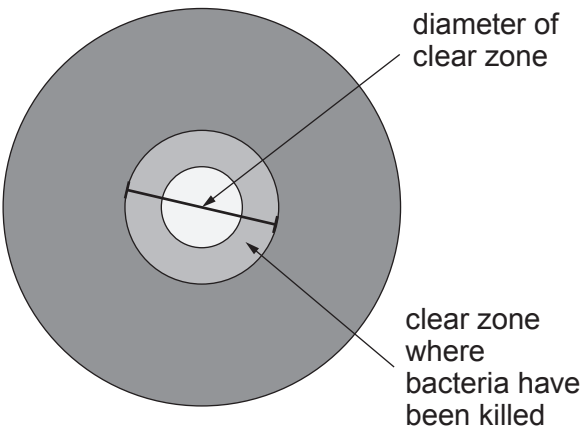
6. Mair investigated four antibiotics (**A**, **B**, **C** and **D**) to compare how effective they were at killing bacteria.
- She set up four agar plates with bacteria growing on them.
 - She placed a filter paper disc containing a different antibiotic in the centre of each plate.
 - She put the plates in an incubator.
 - After three days, she measured the diameter of the clear zone where bacteria had been killed.
 - She calculated the area of the clear zone.

Image 6.1

Agar plate at start of the investigation



Agar plate at three days



- (a) Mair used aseptic techniques to set up the agar plates.
- (i) Explain why it was necessary for her to use **sterilised** forceps to place the filter paper discs on the agar. [1]
-
-
- (ii) Describe how she would have made sure that no bacteria from the environment could enter the agar plates after the filter paper discs were placed on the agar. [1]
-
-
- (b) **Circle** the most suitable temperature for the incubator in a **school** laboratory from the choice below. [1]

10 °C 25 °C 37 °C 60 °C



(c) **Table 6.2** shows Mair’s results.

Table 6.2

	Antibiotic A	Antibiotic B	Antibiotic C	Antibiotic D
Area of clear zone on agar plate (mm ²)	270	360		450

(i) The **radius** of the clear zone for antibiotic **C** was 11 mm.

Complete Table 6.2 by calculating the area of the clear zone for antibiotic **C**, using the formula below, where r = radius.

Area of circle = πr^2 ($\pi = 3.14$)

Give your answer to the **nearest whole number**.
Space for working

[3]

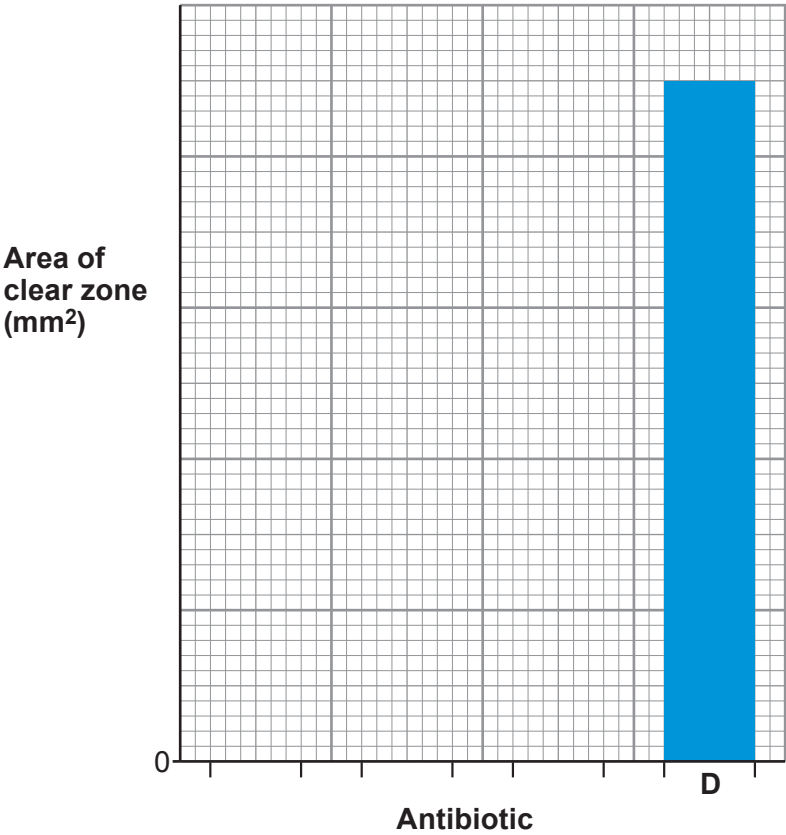
(ii) Use the data in **Table 6.2** to complete **Bar chart 6.3** by:

[3]

- I. adding the scale for the area of the clear zone .
- II. drawing bars for antibiotics **A**, **B** and **C** and labelling your bars.



Bar chart 6.3



(iii) From Bar chart 6.3, state which **one** of the antibiotics was the most effective and explain the reason for your choice. [3]

Antibiotic

Reason

.....
.....

(d) (i) State how Mair could check that her results were reproducible. [1]

.....
.....

(ii) Suggest **one** way in which the investigation could be extended to give more information about the effects of antibiotics on the growth of bacteria. [1]

.....
.....



7. Image 7.1 shows a section through the skin.

(a) Complete labels A and B on Image 7.1.

[2]

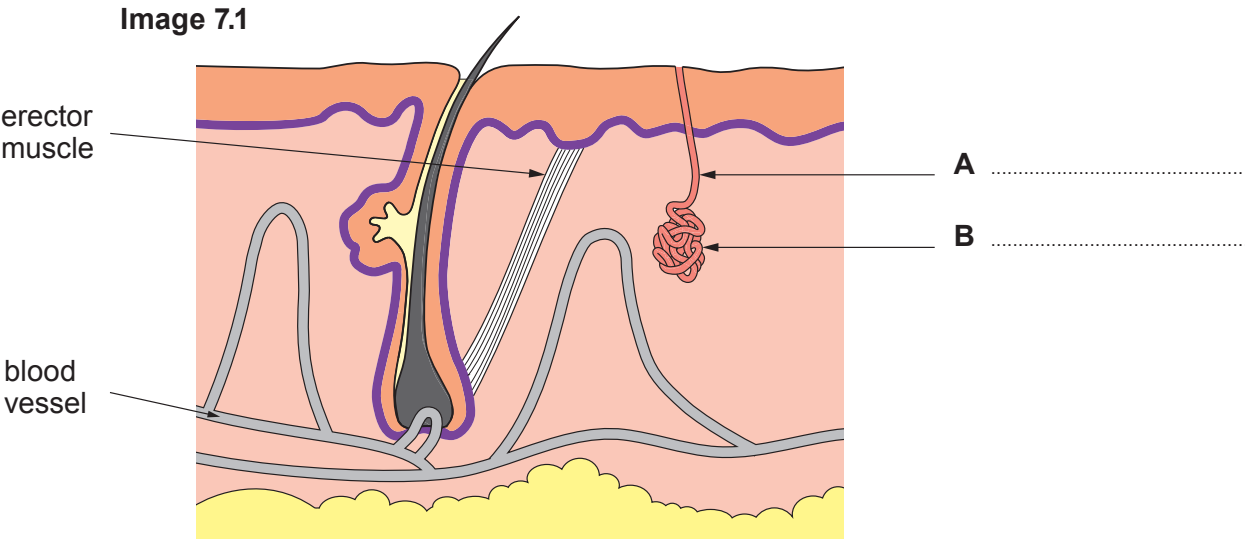


Table 7.2 shows some features of the skin at two air temperatures.

	Air temperature	
	8 °C	28 °C
Volume of sweat produced per day (cm ³)	800	1600
Diameter of blood vessel (mm)	0.005	0.1
Erector muscle	contracted	relaxed

(b) Using the information in Table 7.2, describe the changes in the skin which occur as the air temperature increases. Explain how each of these changes helps to cool the body.

[6 QER]

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Examiner
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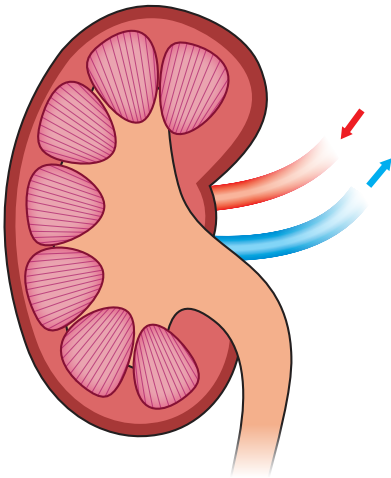
.....

8



8. **Image 8.1** shows a section through a human kidney.

Image 8.1



- (a) **Draw arrows on Image 8.1** to label the following parts: [2]
- (i) the medulla;
 - (ii) the renal artery.

(b) **Table 8.2** shows the results of an investigation on a patient in a hospital.

Table 8.2

Substance	Concentration in blood entering kidney (au)	Concentration in blood leaving kidney (au)	Present in urine (✓ or ✕)
glucose	145	32	✓
protein	120	120	
urea	93	0	

- (i) I. **Complete Table 8.2** to show which substance(s) would be present in the urine of this patient. One row has been done for you. [1]
- II. State which **one** of the substances is a waste product. [1]
-
- III. Doctors suggested that this patient had diabetes. From **Table 8.2**, state the evidence to support this suggestion. [1]
-
-



Examiner
only

(ii) Describe the chemical tests which could be carried out to confirm whether or not the urine contained glucose and protein. For each test state the colours you would expect **for the results shown in Table 8.2.**

I. glucose test [3]

.....

.....

.....

II. protein test [2]

.....

.....

(iii) **Complete the risk assessment** below which shows one hazard linked to these tests. [1]

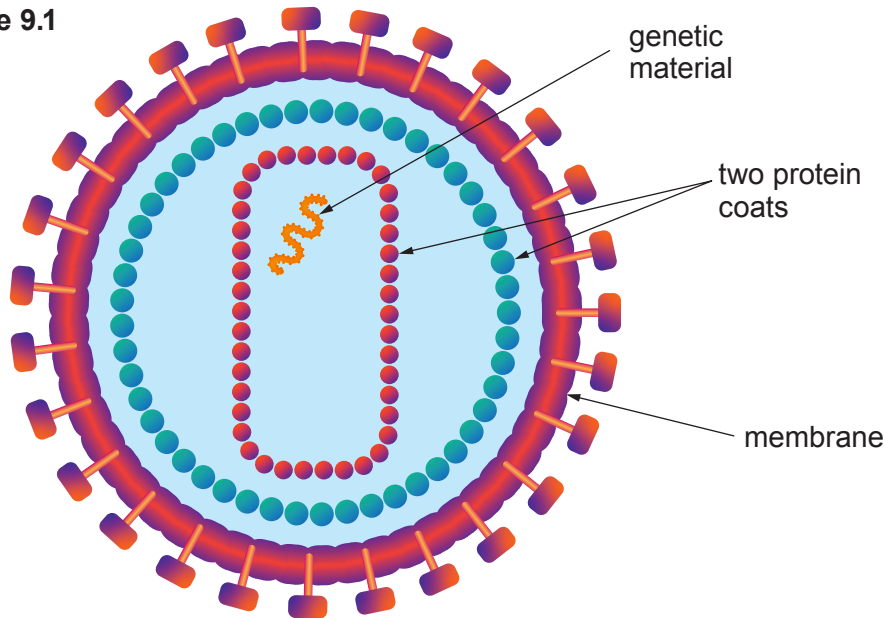
Hazard	Risk	Control measure
Chemical reagent is an irritant		
		
		
		



9. **Image 9.1** shows the structure of the human immuno-deficiency virus (HIV).

In some of the people who have an HIV infection, the virus can lead to the disease AIDS, which can be fatal.

Image 9.1



(a) State **two** ways, visible in **Image 9.1**, in which the structure of HIV is different from other viruses. [1]

- 1.
- 2.

(b) State how HIV is spread from one person to another and **one** measure which can be taken to reduce the spread. [2]

.....

.....

.....

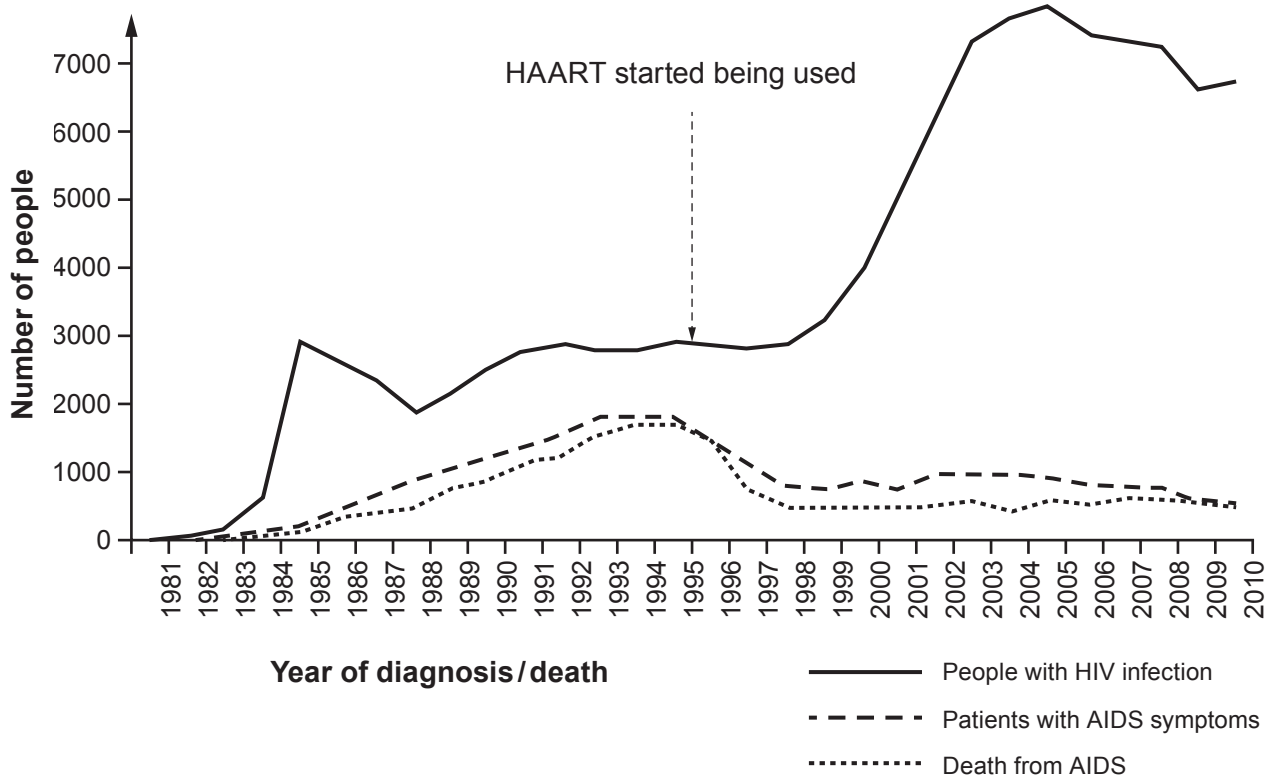


- (c) In 2000, the United Nations set a target that 90% of people with HIV should be receiving treatment. This treatment should be effective at reducing the number of people developing AIDS.

By 2010, 97% of people with HIV in the UK were being treated with the anti-viral drug, HAART.

Graph 9.2 shows the results from investigations on the effectiveness of HAART.

Graph 9.2



Using the information provided on this page, give the evidence that by 2010 the UK had met the UN target by

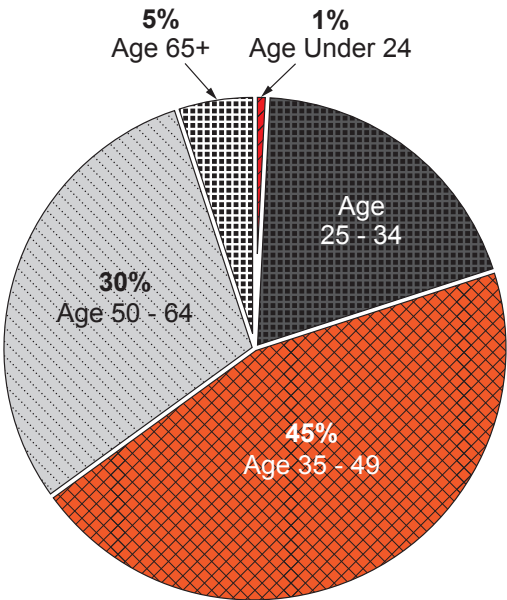
- (i) providing treatment to enough people; [1]

- (ii) providing treatment which was effective. [1]



- (d) **Graph 9.3** shows the percentages of people in different age groups in the UK who were known to be infected with HIV in 2017. The total number of infected people was 101 610.

Graph 9.3



People infected with HIV in the UK

From the information given above calculate

- (i) **the percentage** of people in the 25–34 age group, [1]

..... %

- (ii) **the number** of people in the 35–49 age group. [2]

..... people.

- (iii) Most campaigns which encourage people to take treatment for HIV infections are targeted at those under 35. Some charities, however, say it would be more effective to target campaigns at older age groups.

Using the information in **Graph 9.3**, state **one** piece of evidence which supports the point of view of some charities. [1]

.....

END OF PAPER

